UM-2 solution

An accurate approximate solution of the coupled field equations, with Kohn-Sham theory for the dirac16complex field coupled to gravity and to the Standard-Model fields, three generations of quarks and leptons, and the flavour conversion of the three neutrinos

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Solution, version 3, 2026-10-18 (version 2: 2026-10-18; version 1: 2026-10-17)

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1 How to read this document

This document presents the approximate solution of the coupled field equations of UM-2 and the verification of each step. It is self-contained: every equation it uses is written here. Labels: AUTHOR, RECORD, STANDARD, ASSUMED, INVENTED, DERIVED (computed and verified; the verification is stated), ESTIMATE, OPEN. Units $\hbar = c = 1$; in the numerical work $H = 1$ and $m = H$ unless a physical unit is given. The coordinates are the author's $x_0, x_1, \ldots, x_7$ (x_0 the hidden space direction, x_1, x_2, x_3 space, x_4 the time, x_5, x_6, x_7 the extra times); W^+ carries the electric charge $+e$ and W^- the charge $-e$.

Version 2 (2026-10-18) has three generations of quarks and leptons and three neutrino flavours with their mixing; step 5 is the solution for the flavour conversion of the neutrinos (vacuum, matter, the Sun, accelerators, the early Universe and N_{eff}, the cosmic neutrino background, supernovae). The chiral tip Z portal of version 1 is removed (step 4).

Version 3 (2026-10-18) completes the solution by the author's order to find and document an accurate approximate solution of the coupled field equations with Kohn-Sham theory for dirac16complex coupled to gravity and to every Standard-Model field: step 8 is the coupled solution in ten environments (the Kohn-Sham levels with the mean fields of the Standard-Model fields, the action of dirac16complex back on them, and the response of the block, iterated to self-consistency); step 9 is the thermal history with the x_0 masses and the dark-matter relic from the Boltzmann equation, with the new portal I13 that gives the relic within the unitarity limit; step 10 gives the magnitudes of the terms of the metric power series; step 5 adds the Majorana alternative and step 6 the matter power spectrum (σ_8). After the independent review of version 3 the I13 sum leaves out the direction x_0, for which the bilinear is not Hermitian; the coupled solution of step 8 uses the full adiabatic block response with the hot plasma and the order-α electroweak condensates; step 9 adds the self-interaction channel of the dark matter and corrects the elastic ν_R rate and the fractions of the annihilation rate above the unitarity crossings.

Symbols with more than one use: k is the block variable $1/f^2$ (k_*, δk, $G(k)$), the Lovelock order as the index of α_k, $E_{(k)}$, $\mathcal{L}_{(k)}$ (written p where the block variable appears in the same formula), the momentum of a mode along x_1, x_2, x_3 in h_j, $\vec{k}$, Δk and $\varepsilon = e^{-a_4}|k|$, the comoving Fourier wavenumber in the perturbation equations of step 6, and the wave number of a flavour mode in step 5. $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and Λ_4 the four-dimensional one; in the x_0 eigenvalue problems of step 2, $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue; $\Lambda_\star$ is the scale of the effective interactions. n is a number density, except as a level number ($n\pi$) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell; N is a particle number. The subscript t (p_t, v_t) refers to the extra times x_5, x_6, x_7; $\xi = y/L$ is the scaled hidden coordinate; primes denote $d/d\xi$ in step 2 and $d/d\tau$ (conformal time) in step 6. f is the block scale, f_o an occupation number, f in $f\sigma_8$ the growth rate $d\ln D/d\ln a$ (step 6) and, as a subscript, a fermion multiplet (G_f, q_f, $\tilde{I}_{of}$). ρ is an energy density, except for the electroweak parameter

$\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$ (measured value ρ_0), and θ_W is the angle of the gauge couplings. w denotes the Lovelock weights (w_2, w_3), the weight 1/2 of a fold parity in the Kohn-Sham sums, $\sqrt{c^2-e^2}$ in the fermion mass condition and an equation-of-state parameter. ψ is a Standard-Model fermion field, in step 5 a flavour amplitude and in step 6 the Newtonian potential; σ is a Pauli matrix ($\sigma_1, \sigma_2, \sigma_3$), the annihilation cross-section (σv), σ_{SI} the scattering cross-section, σ_b the cosmic variance of a bin of multipoles and, in step 6, the shear; t_o is an orbital density. Flavour: $i = 1, 2, 3$ is the generation index and labels the neutrino mass states, $\alpha \in \{e, \mu, \tau\}$ a flavour index (a colour index on G^α, the density ratio $\alpha = n_{\bar\nu_e}/n_{\nu_e}$ in the fast-flavour analysis of step 5, and α, α_s the electromagnetic and strong couplings); U is the PMNS and V the CKM matrix (the matter potentials carry subscripts, V_{CC}, V_{NC}, $V_{\nu\nu}$; $V_\perp$ is a volume); κ_ν, κ_e, κ_u, κ_d are the tip matrices; $\mu = \sqrt{2} G_F n_{\nu_e}$ (no subscript) is the unit of the neutrino self-interaction, μ_ψ a bulk mass, μ_N the dark-matter-nucleon reduced mass and μ the muon; e is the elementary charge, the electron, the eigenvalue $\varepsilon f L$, the amplitude of a portal mean field (step 3) and Euler's number (a growth "by e", step 5); J the Jarlskog invariant (and the block operator $\gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$), J_n the moments of the fast-flavour dispersion relation; $G(k)$ is the gravity factor of the block, $G(u)$ the electron-lepton-number distribution of step 5, and G_N, G_F, G_g, G_W, G_B, G_V are couplings; $F(T_1, T_2)$ is the energy-transfer function of the decoupling equations, F_h the decoupling factor of ΔN_{eff} and F in $\bar\chi\chi FF$ a field strength; $H = 14.296$ TeV is the eight-dimensional scale, except in the decoupling estimate of the fold band (step 4), in the temperature equations of step 5 and in the Boltzmann equations of step 9, where H is the Hubble rate; $\eta = HL = 3$ is the depth of the cell, $\eta^{\mu\nu}$ a flat metric; Λ^μ are the first moments of $G(u)$; $s_{ij} = \sin\theta_{ij}$, $c_{ij} = \cos\theta_{ij}$ (c with the subscripts L, Q, R, ψ a localisation); $u = \cos\vartheta$ the direction cosine of a neutrino in step 5 (u also the up quark). Version 3: G_V, G_A and $G = (8G_g^2 + 3G_W^2 + G_B^2)^{1/2}$ are the strengths of I11, I13 and I12 (G_A^{B-L} the strength a $B-L$ variant of I13 would need, G_λ that of the self-interaction channel, G_f the coupling of one fermion multiplet and $G_i = \mathcal{O}\, c_i g_i^2/\Lambda_\star^3$ the I12 strength of the gauge group i); $\tilde I_{of}$, $\tilde I_{of}^{(b)}$, $\tilde I_\lambda$ overlap integrals (s_{DF} a transition scalar density); q_f a charge ($B-L$ for I11, baryon number q_f^B for I13); $P_b = (1-\gamma^{x_0})/2$, $P_a = (1+\gamma^{x_0})/2$; J^B the baryon-number current; c_ρ, c_3, c_p the coefficients of the adiabatic block response, $\bar p = (p_0 + 3p_t)/4$ the hidden pressure and p_2 the order-α pressure of a gauge plasma (step 8); in step 9 $x = M/T$ (M the dark-matter mass, T the temperature), $Y = n/s$ with s the entropy density (in $\sigma(s)$, $\sqrt{s}$ and the thermal averages s is the Mandelstam variable, the squared centre-of-mass energy of a pair), g_ρ, g_s the effective numbers of degrees of freedom ($g_{s,e\gamma}$ that of the photons and $e^\pm$ alone), $\delta = \frac{1}{3}\, d\ln g_s/d\ln T$, γ a rate density (in step 5 γ is the adiabaticity of a resonance), K_1, K_2 modified Bessel functions of the second kind, $\beta = \sqrt{1-4M^2/s}$ the velocity of a dark-matter particle in the centre-of-mass frame (β_W is the tip parameter of the W), $v = 2\beta$ the relative velocity, p the centre-of-mass momentum, J the total angular momentum of a partial wave, N_c the number of Weyl components of a multiplet, which share its overlap (3 for a quark singlet, 6 for a quark doublet, 2 for a lepton doublet, 1 for a charged-lepton singlet), $N_\chi = N_F = 2$ the number of fold bands (the fold zero modes of Ψ_A and Ψ_B, two Dirac fermions) and $\sum_{A,B}$ the sum over the two dark-matter species (the lowest odd states of Ψ_A and Ψ_B); in step 10 Δ_Φ is the rms Newtonian potential per logarithmic wavenumber, α_n the Yukawa strength of a Kaluza-Klein graviton and $a = 3\eta/2$ in its profile (a is otherwise the scale factor and, with b, a component of an orbital).

2 The equations being solved

The action of UM-2 is

$$S = \frac{1}{2\kappa}\int d^8x\sqrt{|g|}\Big(\sum_{k=1}^{3}\alpha_k\mathcal{L}_{(k)} - 2\Lambda\Big) - \int d^8x\sqrt{|g|}\,\rho_b(f) + S_\Psi + S_{gauge} + S_f + S_{tip} + S_{I11} + S_{I12} + S_{I13},$$

with the author's $\alpha_k = w_k H^{2-2k}$, $w = (1, \frac{1}{32}, \frac{1}{1152})$, $\Lambda = -10H^2$; the stabiliser $\kappa\rho_b = \kappa\rho_b^*(f_*/f)^8$; the dirac16complex Lagrangian $\sqrt{|g|}[\frac{1}{2}(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi) - m\bar\Psi\Psi - \frac{\lambda}{2}(\bar\Psi\Psi)^2]$ with the canonical spin connection; the eight-dimensional $SU(3)_c \times SU(2)_L \times U(1)_Y$ gauge fields and three generations of fermions, $\psi_i \in \{Q_i, u_{Ri}, d_{Ri}, L_i, e_{Ri}, \nu_{Ri}\}$ ($i = 1, 2, 3$), with bulk masses $\mu_{\psi i}$ (the doublets of a sector sharing one, $\mu_{Li} = \mu_L$, $\mu_{Qi} = \mu_Q$); the tip sigma field $\Sigma \in SU(2)$ (no Higgs particle) with the tip Yukawa couplings $-\sum_{ij}(\bar L_i\Sigma(\Lambda_\ell)_{ij}\ell_{Rj} + \bar Q_i\Sigma(\Lambda_q)_{ij}q_{Rj} + \text{h.c.})$ of complex 3×3 generation matrices; and the portals

$$S_{I11} = \int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}(i\bar\Psi\gamma^M\Psi)\,\mathcal{J}_M, \qquad \mathcal{J}_M = \sum_{i=1}^{3}\sum_{\psi} q_\psi(i\bar\psi_i\gamma_M\psi_i),$$

$$S_{I12} = -\int d^8x\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\,\mathcal{F}, \qquad \mathcal{F} = c_G G^\alpha_{MN}G^{\alpha MN} + c_W W^a_{MN}W^{aMN} + c_B B_{MN}B^{MN},$$

$$S_{I13} = \int d^8x \sqrt{|g|}\, \frac{c_A}{\Lambda_\star^6} \sum_{M \neq x_0} (i\bar\Psi \gamma^M P_b \Psi)\, J^B_M, \qquad P_b = \frac{1-\gamma^{x_0}}{2}, \qquad J^B_M = \frac{1}{3}\sum_{i=1}^{3} \sum_{\psi \in \{Q_i, u_{Ri}, d_{Ri}\}} i\bar\psi \gamma_M \psi$$

(I11 with the $B-L$ charges $q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_L = q_{e_R} = q_{\nu_R} = -1$; I13 new in version 3, its sum over the seven directions $M = x_1, \ldots, x_7$, for which $i\bar\Psi\gamma^M P_b \Psi$ is Hermitian, while for $M = x_0$ it is not, $(iC\gamma^{x_0} P_b)^\dagger = iC\gamma^{x_0} P_a$ with $P_a = (1+\gamma^{x_0})/2$; DERIVED with the author's matrices). The field equations are

$$\sum_k \alpha_k E_{(k)}{}^M{}_N + \Lambda \delta^M_N = \kappa T^M{}_N, \qquad \gamma^M D_M \Psi = \Big[m + \lambda \bar\Psi \Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - i\frac{c_V}{\Lambda_\star^6} \mathcal{J}_M \gamma^M \Psi - i\frac{c_A}{\Lambda_\star^6} \sum_{M \neq x_0} J^B_M \gamma^M P_b \Psi,$$

together with the Yang-Mills equations with fermion, tip and portal currents, the Dirac equations of the fermions with the tip Yukawa terms, and the boundary conditions at the fold and the tip.

3 The approximation scheme

The solution uses the small parameters of the problem:

- $\epsilon \sim 10^{-5}$: the metric is $g = \bar g + \epsilon h^{(1)} + \ldots$ (order 0: the background; order 1: the CMB, CNB, BAO);
- $(A/H)^2 \ll 1$ after the primordial era: the matter sources are tiny compared with the stabiliser (today 10^{-92}), so the block responds adiabatically and linearly;
- the Standard-Model masses are far below the gap $3H/(2f)$ of the x_0 operators: four-dimensional effective fields;
- dirac16complex is treated by Kohn-Sham density-functional theory on the static block, with the portal couplings as external mean fields (the Standard-Model fields enter through their densities and condensates).

The hierarchy: (1) the background and the primordial era; (2) the x_0 spectra, i.e. all masses and the mixing matrices; (3) the Kohn-Sham states of Ψ coupled to gravity and to the Standard-Model fields, and their back-reaction on the block; (4) the electroweak sector; (5) the flavour conversion of the three neutrinos, N_{eff} and the CNB; (6) the linear cosmology; (7) the dark-matter relic through I12; and in version 3 (8) the coupled solution in environments, iterated to self-consistency; (9) the thermal history and the relic with I13; (10) the magnitudes of the terms of the metric series.

4 Step 1: the background

4.1 The static block

The order-0 metric is

$$ds^2 = e^{2a_4} d\vec x^{\,2} - dx_4^2 + f^2\big[dy^2 - e^{2Hy} d\vec x_t^{\,2}\big], \qquad y = \frac{\ln|\sin 6Hx_0|}{6H}, \quad f = e^{-a_4} b_4^{1/2},$$

periodic in x_0 with period $\pi/(6H)$ (two halves joined at the fold $y=0$, a regular tip at $y=-L$, $\eta = HL = 3$). With $k = 1/f^2$, $A = \dot a_4$, $h = \dot f/f$, a static block ($h=0$, $A=0$) with the stabiliser ($p_3 = -\rho_b$, $p_0 = p_t = \rho_b$) requires (DERIVED)

$$\kappa\rho_b = 10H^2 - 6H^2 k + \tfrac{3}{8}H^2k^2 = 3H^2k - 10H^2 \implies k_* = 12 + \tfrac{4}{3}\sqrt{51} = 21.5219,$$

$$f_* = 0.215556, \qquad G(k_*) = 1 - 24w_2k_* + 72w_3k_*^2 = \tfrac{20}{3} + \sqrt{51} = 13.8081 > 0,$$

$$\kappa\rho_b^* = (26 + 4\sqrt{51})H^2 = 54.566H^2.$$

$G(k_*) > 0$: four-dimensional gravity is attractive with the author's weights and Λ unchanged. (Without the stabiliser the only static block with $A=0$ is $k_0 = 8$ with $G = -1$, repulsive.) The linearised system has the eigenvalues $0, \pm i\omega_r$,

$$\omega_r^2 = \Big(\frac{31552}{2249} - \frac{2416\sqrt{51}}{6747}\Big)H^2, \qquad \omega_r = 3.38705H,$$

so the block is stable; the radion mass is $m_r = 3.387H = 48.42$ TeV. The whole stabiliser family $p_0 = \nu\rho_b$ ($\nu = 1/4$ to 3) gives stable blocks with $\omega_r = 3.33H$ to $4.10H$ (DERIVED).

4.2 The primordial era: the deflating extra times

The full nonlinear equations, i.e. the three independent components of the field equations of the order-0 metric with the author's weights and $\Lambda = -10H^2$ ($A = \dot{a}_4$, $h = \dot{f}/f$, $k = 1/f^2$, dots d/dx_4; $\dot{k} = -2kh$; the source the stabiliser, $\kappa\rho_b = \kappa\rho_b^*(k/k_*)^4$, $p_3 = -\rho_b$, $\bar{p} = (p_0 + 3p_t)/4 = \rho_b$, plus radiation where stated),

$$
\begin{aligned}
x_4: \quad & -\frac{15A^2h^4}{16H^4} - \frac{5A^3h^3}{4H^4} - \frac{3h^4}{8H^2} - \frac{9Ah^3}{2H^2} - \frac{27A^2h^2}{4H^2} + \frac{9kA^2h^2}{8H^2} \\
& - \frac{3hA^3}{2H^2} + \frac{3hkA^3}{4H^2} - 6h^2 + \frac{3kh^2}{4} - 12Ah + \frac{9Ahk}{2} \\
& - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 + 6kH^2 - \frac{3H^2k^2}{8} \\
& = -\kappa\rho \\
\\
x_1: \quad & \Big[-\frac{h^4}{8H^4} - \frac{Ah^3}{2H^4} - \frac{3h^2}{2H^2} + \frac{kh^2}{4H^2} - \frac{Ah}{H^2} + \frac{Ahk}{2H^2} \\
& \quad - 2 + \frac{3k}{2} - \frac{k^2}{8} \Big] \dot{A} \\
& + \Big[-\frac{Ah^3}{2H^4} - \frac{3A^2h^2}{4H^4} - \frac{3h^2}{2H^2} - \frac{3Ah}{H^2} + \frac{Ahk}{2H^2} - \frac{A^2}{2H^2} \\
& \quad + \frac{kA^2}{4H^2} - 4 + \frac{3k}{2} \Big] \dot{h} \\
& - \frac{Ah^5}{2H^4} - \frac{19A^2h^4}{16H^4} - \frac{A^3h^3}{2H^4} - \frac{15h^4}{8H^2} - \frac{6Ah^3}{H^2} + \frac{Akh^3}{2H^2} \\
& - \frac{17A^2h^2}{4H^2} + \frac{7kA^2h^2}{8H^2} - \frac{hA^3}{H^2} + \frac{hkA^3}{2H^2} - 10h^2 + \frac{9kh^2}{4} \\
& - 8Ah + 3Ahk - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 \\
& + 6kH^2 - \frac{3H^2k^2}{8} \\
& = \kappa p_3 \\
\\
x_0: \quad & \Big[-\frac{3Ah^3}{8H^4} - \frac{9A^2h^2}{16H^4} - \frac{9h^2}{8H^2} - \frac{9Ah}{4H^2} + \frac{3Ahk}{8H^2} - \frac{3A^2}{8H^2} \\
& \quad + \frac{3kA^2}{16H^2} - 3 + \frac{9k}{8} \Big] \dot{A} \\
& + \Big[-\frac{9A^2h^2}{16H^4} - \frac{3hA^3}{8H^4} - \frac{3h^2}{8H^2} - \frac{9Ah}{4H^2} - \frac{9A^2}{8H^2} + \frac{3kA^2}{16H^2} \\
& \quad - 3 + \frac{3k}{8} \Big] \dot{h} \\
& - \frac{9A^2h^4}{16H^4} - \frac{17A^3h^3}{16H^4} - \frac{9A^4h^2}{16H^4} - \frac{3h^4}{8H^2} - \frac{27Ah^3}{8H^2} - \frac{45A^2h^2}{8H^2} \\
& + \frac{3kA^2h^2}{16H^2} - \frac{27hA^3}{8H^2} + \frac{9hkA^3}{16H^2} - \frac{3A^4}{8H^2} + \frac{3kA^4}{16H^2} - 6h^2 \\
& + \frac{3kh^2}{8} - 9Ah + \frac{9Ahk}{8} - 6A^2 + \frac{9kA^2}{4} - 10H^2 \\
& + 3kH^2 \\
& = \kappa\bar{p}
\end{aligned}
$$

(DERIVED, sympy), were integrated (the x_4 constraint preserved to 6×10^{-10}) from an expanding 3-space and a block larger than the stabilised one ($k_0 < k_*$) that deflates at $h_0 = -H/2, -H/3, -H/10$, for k_0 from 1.5 to 20 (DERIVED):

start k_0	$h_0 = -H/10$	$h_0 = -H/3$	$h_0 = -H/2$
1.5	singular at $x_4 = 1.44/H$	singular at $1.26/H$	singular at $1.14/H$
2.3	singular at $0.33/H$	singular at $0.31/H$	no expanding start
2.7	$k(60/H) = 21.415$	21.435	21.451
3	21.412	21.434	21.455
6	21.405	21.453	21.500
10	21.446	21.541	21.584
20	21.634	21.640	21.638

Every start above the separatrix $k_c = 2.5301, 2.5991, 2.6983$ (for $h_0 = -H/10, -H/3, -H/2$; just above the unstable static block $k_- = 2.4781$) ends at the stable block: the extra times x_5, x_6, x_7 and the hidden direction x_0 deflate at rates up to $|h| = 2.63H$ (in the stored runs $k_0 = 3, 5, 10$), by up to $f(k_c)/f_* = 2.917$, reach k_* within $x_4 < 1/H$, and then oscillate about it with the radion frequency ($3.3868H$ measured, $3.3870H$ linear), while 3-space keeps expanding; the oscillation energy redshifts like dust ($A\,x_4 \to 0.667$ to 0.670; 2/3 for dust). The extra times therefore deflate exponentially only in a short primordial era and are then frozen: the masses, all of which depend on f, become constant.

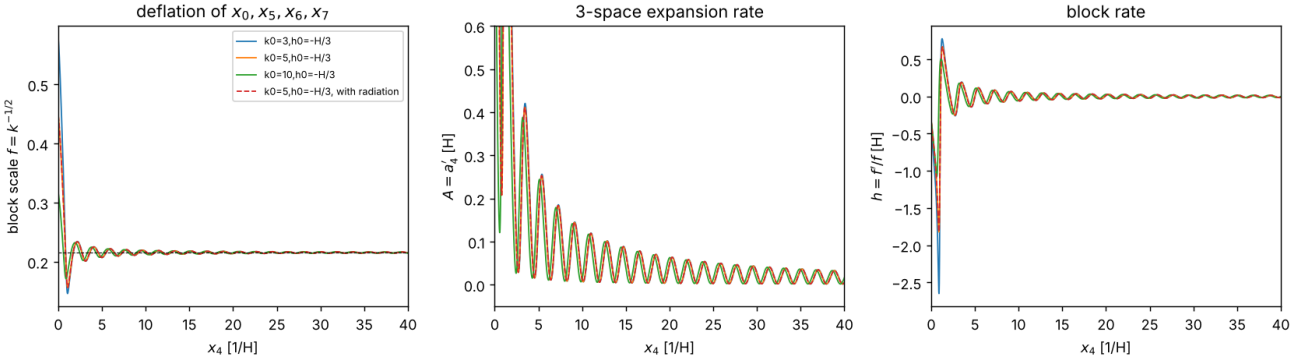


Figure 1: The primordial era: the block scale f, the 3-space expansion rate A and the block rate h for starts $k_0 = 3, 5, 10$ at $h_0 = -H/3$ (one with radiation), from the full nonlinear Einstein-Lovelock equations with the author's weights.

4.3 The response of the block to matter, and four-dimensional gravity

To first order about the static block with block-averaged matter sources ($\bar{p} = (p_0 + 3p_t)/4$; A^2 of first order, $3G(k_*)A^2 = \kappa\delta\rho$) (DERIVED):

$$\delta\ddot{k} + \omega_r^2\,\delta k = -2k_*\big(s_{h3}\,\kappa\delta p_3 + s_{hp}\,\kappa\delta\bar{p} + q_h\,\kappa\delta\rho\big), \qquad s_{h3} = 0.028666, \quad s_{hp} = 0.037320, \quad q_h = -0.009555;$$

the adiabatic shift is $\delta k = (-0.107556\,\kappa\delta p_3 - 0.140028\,\kappa\delta\bar{p} + 0.035852\,\kappa\delta\rho)/H^2$ (radiation, $\delta p_3 = \delta\rho/3$, drops out exactly: $0.107556 = 3 \times 0.035852$), and 3-space obeys $\delta\dot{A} = -\kappa(\delta\rho + \delta p_3)/(2G(k_*))$. Together with the constraint $3G(k_*)A^2 = \kappa\rho$ and the dimensional reduction ($\sum_p \alpha_p \mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_b = G(k)R^{(4)} + \text{topological} - 2G(k)\Lambda_4$, p the Lovelock order), the four-dimensional gravity of UM-2 is general relativity,

$$G^{(4)}_{\mu\nu} + \Lambda_4 g_{\mu\nu} = 8\pi G_N T^{(4)}_{\mu\nu}, \qquad 8\pi G_N = \frac{\kappa}{G(k_*)V_\perp}, \qquad V_\perp = \frac{2f_*^4 v_t(1 - e^{-3\eta})}{3H},$$

with $\Lambda_4 = [6H^2k(1 - 2w_2k) + \Lambda + \kappa\rho_b]/G(k) = 0$ at the static point; the observed dark energy needs $\Lambda_4/H^2 \sim (H_0/H)^2 = 1.0 \times 10^{-92}$ (tuned; OPEN).

5 Step 2: the masses from the x_0 dependence of the wave functions

Every four-dimensional mass is an eigenvalue of the x_0 operator of its field on the static block; the eigenfunctions depend on x_0 through $y(6Hx_0)$ and are periodic with period $\pi/(6H)$ (the author's requirement). With $\xi = y/L$:

5.1 Gauge bosons and the graviton

$$-e^{-3\eta\xi}\big(e^{3\eta\xi}u'\big)' = \Lambda u, \qquad \Lambda = (m\,fL)^2, \qquad u'(0) = 0 \text{ (even) or } u(0) = 0 \text{ (odd)}, \qquad u'(-1) = \beta u(-1),$$

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0, \qquad q^2 = \Lambda - \frac{9\eta^2}{4}.$$

mode ($\eta = 3$)	$\Lambda = (m\,fL)^2$	status
photon, gluons, graviton ($\beta = 0$, flat profile)	0 exactly	DERIVED
first even Kaluza-Klein level	$9\eta^2/4 + \pi^2 = 30.1196$	DERIVED (finite elements 1.9×10^{-10})
first odd level	27.0890 ($q\cot q = -9/2$)	DERIVED (finite elements 4.5×10^{-10})
gap	$9\eta^2/4 = 20.25$	DERIVED
W^+, W^- ($\beta_W = 0.011912$)	1.32149×10^{-5}	DERIVED (fit, finite elements 3×10^{-14})
Z ($\beta_Z = \beta_W/\cos^2\theta_W = 0.015343$, angle of the couplings $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$)	1.70150×10^{-5}	DERIVED
Dirichlet bound ($\beta \to \infty$)	0.010016: $m_W \le 0.01824\, m_{KK}$	DERIVED

Here θ_W is the angle of the gauge couplings, $\cos^2\theta_W = \hat{g}^2/(\hat{g}^2 + \hat{g}'^2)$. The bending of the tip-localised profiles moves the mass ratio $m_W^2/m_Z^2 = \Lambda_W/\Lambda_Z$ away from $\cos^2\theta_W$, and the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z\cos^2\theta_W)$ exceeds 1 ($\rho - 1 = 3.2 \times 10^{-8}$ at $\beta_W = 10^{-6}$: custodial symmetry at leading order; 6.4×10^{-5} at $\beta_W = 0.002$, 3.2×10^{-3} at 0.1). This tree-level custodial violation is fitted to the measured $\rho_0 - 1 = 0.00038$ (the violation beyond the Standard-Model loops), and the angle of the couplings is fixed by the measured masses, $\rho = m_W^2/(m_Z^2\cos^2\theta_W) = \rho_0$, i.e. $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$ (on-shell $m_W^2/m_Z^2 = 0.776663$); both masses are then reproduced. With $m_W = 80.3625$ GeV (STANDARD, PDG 2026) this gives the scales (DERIVED)

quantity	value
Kaluza-Klein mass $m_{KK} = m_W\sqrt{\Lambda_1/\Lambda_W}$	121.32 TeV
$H = m_{KK} f_* \eta/\sqrt{\Lambda_1}$	14.296 TeV
proper half cell fL; period of x_0, $\pi/(6H)$	8.926×10^{-21} m; 7.227×10^{-21} m
gap $3H/(2f_*)$	99.48 TeV
radion mass $3.387H$	48.42 TeV
model $m_Z = m_W\sqrt{\Lambda_Z/\Lambda_W}$	91.1879 GeV (the measured value, reproduced)

5.2 Fermions: masses and mixing of three generations

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + \mu\,\sigma_2 + e^{-a_4}k\,\sigma_3\Big],$$

with k the momentum of the mode along x_1 (not the block variable) and $c = f\mu L$. In $\xi = y/L$, with $e = \varepsilon fL$, the $k = 0$ components (a, b) of a block $j = +1$ obey $a' = ca - eb$, $b' = ea - cb$ (primes $d/d\xi$). In the selected blocks of the Standard-Model fields, $P_{sel} = \frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-i\gamma^{x_5}\gamma^{x_6}}{2}$, the fold conditions $P_{sel}(1 \mp \gamma^{x_0})\psi = 0$ ($\gamma^{x_0} = \sigma_3$ in block form) are $b(0) = 0$ for a doublet ($c_L > 0$) and $a(0) = 0$ for a singlet ($c_R < 0$), so that

$(a_L, b_L) = (\cosh w\xi + (c_L/w)\sinh w\xi,\ (e/w)\sinh w\xi)$ and $(a_R, b_R) = (-(e/w)\sinh w\xi,\ \cosh w\xi - (c_R/w)\sinh w\xi)$, and at the tip $\xi = -1$ $(a_L, b_L) = (C_L - c_L s_L, -s_L e)$ and $(a_R, b_R) = (s_R e, C_R + c_R s_R)$, $s = \sinh w/w$, $C = \cosh w$, $w = \sqrt{c^2 - e^2}$; with the tip mixing $b_L = \kappa_f b_R$, $a_R = -\kappa_f a_L$ of one generation the mass is the lowest root of

$$D(e) = e^2 s_L s_R - \kappa_f^2 (C_L - c_L s_L)(C_R + c_R s_R) = 0, \qquad e \simeq \kappa_f \zeta(c_L)\zeta(-c_R), \quad \zeta(c)^2 = \frac{c\, e^{-c}}{\sinh c}.$$

For three generations the tip conditions $b_L = \kappa b_R$, $a_R = -\kappa^\dagger a_L$ (vectors over the generations, κ the 3×3 tip matrix of a sector) give, with the doublet amplitudes $\mathcal{A}$ and the singlet amplitudes $\mathcal{B}$,

$$\Big[\mathrm{diag}(b_{L\alpha}) + \kappa\, \mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\kappa^\dagger\, \mathrm{diag}(a_{L\alpha})\Big]\mathcal{A} = 0,$$

and for a common doublet bulk mass and a unitary κ the determinant is $\prod_i (b_L a_{Ri} + a_L b_{Ri})/a_{Ri}$ with $b_L a_R + a_L b_R = -D(e)|_{\kappa_f = 1}$, the null vectors are $\mathcal{A}^{(i)} = \kappa \hat{e}_i$ and $\mathcal{B}^{(i)} \propto \hat{e}_i$: the masses are the one-generation roots with $\kappa_f = 1$ and the left-handed rotation of each sector is its tip matrix, so that $U = \kappa_e^\dagger \kappa_\nu$ and $V = \kappa_u^\dagger \kappa_d$ (DERIVED, `um2_flavour.py`, mpmath with 50 digits: the roots of the 3×3 determinant agree with the one-generation roots, both refined in the same computation, to 3.0×10^{-42} in the four sectors (the masses stored by the mass computation agree with them to 4.5×10^{-11}), with $\kappa_\nu = U_{PMNS}$, $\kappa_e = 1$, $\kappa_u = 1$, $\kappa_d = V_{CKM}$, and the null vectors with the columns of κ to 1.1×10^{-16}; for a random complex non-unitary tip matrix the exact roots agree with the singular values of $\zeta(c_L)\,\kappa\,\mathrm{diag}(\zeta(-c_{Ri}))$ to 3.5×10^{-5}, 2.8×10^{-6}, 2.0×10^{-6} and the null vectors with its left singular vectors).

With one doublet bulk mass per sector, fixed by the heaviest member with $c_R = -c_L$ (τ, t), the masses require (DERIVED, `um2_spectra.py`; exact roots; direct shooting of the coupled (L, R) Dirac system agrees with the determinant to 6.1×10^{-10} (e), 1.5×10^{-11} (τ), 5.6×10^{-12} (t); the first-order formula agrees with the exact root to the relative accuracy in the last column):

fermion	mass	doublet c	singlet c_R	μ_R/H	singlet width $fL/(2\lvert c_R\rvert)$	$e = m\,fL$	first order
ν_1	1.000 meV	5.95288	-35.04518	-54.194	1.27×10^{-22} m	4.5235×10^{-17}	5.7×10^{-7}
ν_2	8.712 meV	5.95288	-32.84810	-50.796	1.36×10^{-22} m	3.9409×10^{-16}	6.1×10^{-7}
ν_3	50.140 meV	5.95288	-31.07017	-48.047	1.44×10^{-22} m	2.2681×10^{-15}	6.5×10^{-7}
e	0.51100 MeV	5.95288	-14.55391	-22.506	3.07×10^{-22} m	2.3115×10^{-8}	1.4×10^{-6}
μ	105.658 MeV	5.95288	-8.98093	-13.888	4.97×10^{-22} m	4.7795×10^{-6}	2.2×10^{-6}
τ	1776.93 MeV	5.95288	-5.95288	-9.205	7.50×10^{-22} m	8.0380×10^{-5}	6.8×10^{-6}
u	2.16 MeV	3.38230	-15.43001	-23.861	2.89×10^{-22} m	9.7708×10^{-8}	1.3×10^{-4}
c	1.2730 GeV	3.38230	-8.76830	-13.559	5.09×10^{-22} m	5.7585×10^{-5}	2.2×10^{-4}
t	172.56 GeV	3.38230	-3.38230	-5.230	1.32×10^{-21} m	7.8058×10^{-3}	1.1×10^{-3}
d	4.70 MeV	3.38230	-14.62578	-22.617	3.05×10^{-22} m	2.1261×10^{-7}	1.3×10^{-4}
s	93.5 MeV	3.38230	-11.51582	-17.808	3.88×10^{-22} m	4.2295×10^{-6}	1.7×10^{-4}
b	4.183 GeV	3.38230	-7.50053	-11.599	5.95×10^{-22} m	1.8922×10^{-4}	2.6×10^{-4}

(doublets: $\mu_L = 9.2055\,H$, $\mu_Q = 5.2304\,H$; the masses are the PDG values, the light-quark masses in $\overline{MS}$ at 2 GeV, $m_c(m_c)$, $m_b(m_b)$ and the directly measured m_t, used as tree-level targets; m_1 is ASSUMED, m_2, m_3 follow from the NuFIT 6.0 splittings of step 5; the masses are STANDARD inputs, PDG 2025.) The mixing matrices are $U = R_{23}U_{13}(\delta)R_{12}$ in the standard form

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}, \qquad J = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta,$$

with the NuFIT 6.0 values ($\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta = 212°$; $J = -0.01777$, $J_{max} = 0.0335$ from these best-fit angles, against NuFIT's marginalised 0.0333 ± 0.0007) for the PMNS matrix and the PDG values ($s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$; $J = 3.117 \times 10^{-5}$) for the CKM matrix, both unitary to 1.1×10^{-16} (DERIVED). The first even Kaluza-Klein gauge mode couples to the zero modes with the overlaps 0.14434 (every left-handed lepton) and −0.30066 (every left-handed quark) relative to the zero mode, and in the mass basis its largest off-diagonal coupling is 1.0×10^{-17}: no tree-level flavour-changing neutral current (with the three charged-lepton $|c_R|$ used as different doublet localisations it would be 0.033). $\mathrm{BR}(\mu \to e\gamma) = \frac{3\alpha}{32\pi}\left|\sum_i U_{\mu i}^* U_{ei} m_i^2/m_W^2\right|^2 = 2.8 \times 10^{-55}$ (DERIVED).

5.3 dirac16complex

With $M = m$, fold condition $\chi_2(0) = 0$ (even) or $\chi_1(0) = 0$ (odd), tip $\chi_2(-L) = 0$ and $c_\psi = f_* mL = 0.646667$, the exact $k = 0$ spectrum is: the fold zero mode ($\varepsilon = 0$, all eight blocks); even bulk levels $\sqrt{m^2 + (n\pi/(fL))^2}$, the lowest $4.959981\,m$; odd levels $\sqrt{m^2 + p^2}$ with $\tan(p\,fL) = -p/m$, the lowest $3.102204\,m$ ($p = 2.93661\,m$, $p\,fL = 1.89901$); and the fold band $\varepsilon = e^{-a_4}|k|$ (massless, $\varepsilon/|k| = 1$ to 6×10^{-14}) (DERIVED).

6 Step 3: Kohn-Sham states of dirac16complex coupled to all fields

6.1 The Kohn-Sham equations

$$h_j\chi_o = \varepsilon_o\chi_o, \qquad M_{eff} = m + \frac{15}{16}\lambda S + \frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3}, \qquad v = -\frac{\lambda}{16}n + \frac{c_V}{\Lambda_\star^6}\sum_{i=1}^{3}\sum_{\psi} q_\psi n_{\psi i},$$

$$n = \sum wgf_o n_o, \quad S = \sum wgf_o s_o, \quad n_o = P\chi_o^\dagger\chi_o, \quad s_o = P\chi_o^\dagger j\sigma_2\chi_o, \quad t_o = P\chi_o^\dagger j\sigma_3\chi_o, \quad P = \frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

$$E_{KS} = \sum gf_o\varepsilon_o - 2\mathrm{Vol}_7\int e^{3Hy}e_{int}dy, \qquad e_{int} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2,$$

$$\rho = \sum wgf_o\varepsilon_o n_o - e_{int}, \qquad p_3 = \sum wgf_o\tfrac{e^{-a_4|k|}}{3}t_o + e_{int}, \qquad p_t = e_{int},$$

$$p_0 = \sum wgf_o[(\varepsilon_o - v)n_o - M_{eff}s_o - e^{-a_4|k|}t_o] + e_{int},$$

with $w = 1/2$ per fold parity, $g = 4r_3(|\vec{n}|^2)$ for the momentum shell $\vec{k} = \Delta k\,\vec{n}$ ($\vec{n} \in \mathbb{Z}^3$, $r_3(|\vec{n}|^2)$ the number of representations of $|\vec{n}|^2$ as a sum of three squares, $\Delta k = 0.25$; here k is the momentum, not the block variable), occupations f_o, and $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 v_t f_*^4 = 34.27$ ($v_t = 1$, $a_4 = 0$; the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - e^{-3\eta})/(3H)$). The orbitals are expanded in the exact free eigenfunctions of each sector (60, 120, 240 per sector), iterated to self-consistency (tolerance 10^{-13}, safeguarded Anderson mixing), and extrapolated in the basis size (Richardson).

6.2 Validation

The independent solver reproduces the record's two reference states (at $f = 1$, $a_4 = 0$): $N = 8$, $\lambda = 157.7$: $E_{KS} = -0.037786682923$ (reference -0.037786682923), HOMO and LUMO to 10^{-14}, integrated $p_0 = -0.08860537545$ (reference -0.08860537548); $N = 592$, $\lambda = 2.131$: $E_{KS} = 358.30104644787$ (identical), integrated $p_0 = -6.7691415097$ (reference -6.7691415108) (DERIVED).

6.3 Ground states on the stabilised block

Two families of states (the eight $k = 0$ fold zero modes, $N = 8$; a fold-band gas, $N = 592$) with $\lambda = 0$, the benchmark $\pm\lambda_B$ ($\lambda_B = 7 \times 10^{-5} H^{-6}$: mean fields below 5 % of m) and the record's coupling scaled to the block (stress test, $\lambda = 157.7 f_*^4 = 0.3405$ and $2.131 f_*^4 = 0.00460$) (DERIVED; 120 functions per sector, relative Richardson errors below 2.2×10^{-9}):

state	E_{KS}	E_{int}	HOMO	LUMO	iterations
$N = 8$, $\lambda = 0$	0	0	0	0.25	1
$N = 8$, $+\lambda_B$	-2.053×10^{-4}	-2.053×10^{-4}	-5.13×10^{-5}	0.24995	14
$N = 8$, $-\lambda_B$	2.053×10^{-4}	2.053×10^{-4}	5.13×10^{-5}	0.25005	14
$N = 8$, stress	-0.70123	-0.42170	-0.14037	0.10859	61
$N = 592$, $\lambda = 0$	361.11614	0	0.79057	0.82916	1
$N = 592$, $+\lambda_B$	359.99060	-1.12646	0.78676	0.82535	14
$N = 592$, $-\lambda_B$	362.23906	1.12121	0.79436	0.83295	13
$N = 592$, stress	307.67139	-32.2217	0.64506	0.68347	56

state	$\int p_3$	$\int p_t$	$\int p_0$	max $\lvert\delta M\rvert$	max $\lvert v\rvert$
$N = 8$, $+\lambda_B$	-2.05×10^{-4}	-2.05×10^{-4}	-4.447×10^{-4}	2.4×10^{-7}	6.7×10^{-4}
$N = 8$, $-\lambda_B$	2.05×10^{-4}	2.05×10^{-4}	4.447×10^{-4}	2.4×10^{-7}	6.7×10^{-4}
$N = 8$, stress	-0.4217	-0.4217	-1.30076	1.98	1.83
$N = 592$, $\lambda = 0$	120.372	0	8.6×10^{-12}	0	0
$N = 592$, $+\lambda_B$	119.245	-1.1265	-2.43688	1.3×10^{-3}	0.050
$N = 592$, $-\lambda_B$	121.493	1.1212	2.43174	1.3×10^{-3}	0.050
$N = 592$, stress	87.614	-32.222	-97.8188	2.09	1.88

(The integrals are $2\mathrm{Vol}_7 \int e^{3Hy}(\cdot)\, dy$; the $N = 8$, $\lambda = 0$ state has all of them zero.)

Checks (DERIVED): $E_{KS} = 2\mathrm{Vol}_7 \int e^{3Hy} \rho\, dy$ to 4×10^{-15}; the y-conservation law $[e^{3Hy} p_0]_{-L}^{0} = 3H \int e^{3Hy} p_t dy$ to 9×10^{-8} (Richardson; the raw residual falls like n_b^{-3}); observed basis orders 4.95 (energy) and 2.97 (hidden pressure); the free gas has $w = \int p_3 / \int \rho = 1/3$ exactly (four-dimensional radiation). The self-interaction shifts the fold-band levels almost rigidly (the spacing between $k = 0$ and the first shell is 0.25 free, 0.25000 at λ_B, 0.2490 in the $N = 8$ stress state). A scalar mean field cannot gap the fold band at $k = 0$: $\chi = (e^{f \int M_{eff} dy}, 0)$ solves $j[-\frac{i}{f}\sigma_1 \partial_y + M_{eff}\sigma_2]\chi = 0$ with both boundary conditions for every $M_{eff}(y)$.

6.4 Dark-matter quanta

Eight quanta in the lowest odd level at $k = 0$ on the $N = 8$ ground state add exactly $8\varepsilon_{DM} = 24.817633063$ (computed 24.817633063) at $\lambda = 0$ and 24.81833 at $+\lambda_B$; they are cold, $w = p_3/\rho = 0$ (2.8×10^{-5} with λ_B); their hidden pressure is $p_0/\rho = 0.772034343$, equal to $-d \ln \varepsilon_{DM} / d \ln f = 0.772034343$ obtained independently from the x_0 eigenvalue (DERIVED): the dark-matter mass is an x_0 eigenvalue, and it pushes on the block.

6.5 The Standard-Model fields acting on dirac16complex

The portal mean fields were applied with amplitude e and shapes given by the sources: for I11 the proper densities of the zero modes times $P = e^{-3Hy}/\mathrm{Vol}_7$, with the density of an unpolarised fermion the average of its doublet and singlet zero-mode densities, $\varrho_c(y) = (2c/L)\, e^{2cy/L}/(1 - e^{-2c})$ for the localisation c: the electron $\frac{1}{2}(\varrho_{c_L} + \varrho_{-c_R(e)})$ ($c_L = 5.9529$, $-c_R(e) = 14.5539$), the up quark $\frac{1}{2}(\varrho_{c_Q} + \varrho_{-c_R(u)})$ ($c_Q = 3.3823$, $-c_R(u) = 15.4300$) and the neutrinos, produced left-chiral, ϱ_{c_L}; for I12 a uniform gluon condensate. Linear responses of the lowest odd level (DERIVED; equal to the Hellmann-Feynman expectation $d\varepsilon/de = \langle\chi|\partial h/\partial e|\chi\rangle$ to 10^{-8}):

portal mean field	$d\varepsilon_{odd}/de$	change of the fold-band gap
I11, electron density (v)	0.824779	5.4×10^{-11}
I11, up-quark density (v)	4.328803	3.1×10^{-9}
I11, neutrino density (v, lepton doublet zero mode)	1.252406	1.1×10^{-10}
I12, gluon condensate (uniform M)	0.707196	2.9×10^{-11}

The broad quark doublet zero mode ($c_Q = 3.38$) overlaps the dark-matter orbital most. The fold band stays massless under every portal. Every orbital has $\chi_2(-L) = 0$, so the scalar density $\bar\Psi\Psi = 2s_2ab$ vanishes at the tip: no tip operator built on $\bar\Psi\Psi$ reaches dirac16complex; the upper component does not vanish there, so a chiral tip current would reach Ψ_A (the projector $P_+ = (1 - B\gamma^{x_0})/2$ annihilates every Ψ_B orbital at the tip), but UM-2 contains none (step 4). The I12 response defines the four-dimensional coefficient of the dark-matter operator: $G_i = \mathcal{O}\, c_i g_i^2/\Lambda_\star^3$ with $\mathcal{O} = 0.7072$.

6.6 Back-reaction on the block

A Kohn-Sham state with the block-averaged sources ρ, p_3 and the hidden pressure $\bar p = (p_0 + 3p_t)/4$ displaces the block by the adiabatic response of step 1,

$$\delta k = \big(0.035852\,\kappa\rho - 0.107556\,\kappa p_3 - 0.140028\,\kappa\bar p\big)/H^2 = \big(0.035852\,\kappa(\rho - 3p_3) - 0.140028\,\kappa\bar p\big)/H^2$$

(the hidden-pressure part is $\delta k = \kappa\bar p/(3H^2 - 4\kappa\rho_b^*/k_*)$, stiffness $-7.1414H^2$), without changing the four-dimensional cosmological term at linear order ($\frac{d}{dk}[6H^2k(1 - 2w_2k) + \kappa\rho_b] = -6 \times 10^{-12}$ at k_*). Cell averages (units of the Kohn-Sham calculation, $H = 1$): $\rho - 3p_3 = 1.80 \times 10^{-5}$ and $\bar p = -1.16 \times 10^{-5}$ ($N = 8$, $+\lambda_B$; $\delta k = 2.27 \times 10^{-6}\kappa$), 0.0987 and -0.0636 ($N = 592$, $+\lambda_B$; $\delta k = 0.0125\kappa$), 0 and 0 (free gas, $w = 1/3$; $\delta k = -5 \times 10^{-13}\kappa$, the numerical zero) (DERIVED); the shift is negligible for any realistic density (relative size $(A/H)^2$). The difference $p_0 - p_t$ (-0.0574 for $N = 592$, $+\lambda_B$) drives a block-shape mode outside the order-0 metric (OPEN).

7 Step 4: the electroweak sector

Charges and couplings (DERIVED): in every generation $Q = T_3 + Y$ gives the up-type quarks u, c, t the charge $2/3$, the down-type quarks d, s, b the charge $-1/3$, the neutrinos 0 and the charged leptons e, μ, τ the charge -1, for both chiralities; $[T_3, T_\pm] = \pm T_\pm$ with $Y(W) = 0$: $W^+ = (W^1 - iW^2)/\sqrt{2}$ **carries** $+e$**,** $W^- = (W^1 + iW^2)/\sqrt{2}$ **carries** $-e$, and W^- is the antiparticle of W^+; the photon couples with $e = gg'/\sqrt{g^2 + g'^2}$ times Q, the Z with $\sqrt{g^2 + g'^2}(T_3 - Q\sin^2\theta_W)$.

Anomalies (DERIVED, exact fractions). With every zero mode written as a left-handed Weyl field ($-Y$ for a right-handed one), n_c colours and n_w weak components, one generation gives

$$\mathcal{A}_{SU(3)^2U(1)} = \sum_{\text{triplets}} n_w Y = 0, \qquad \mathcal{A}_{SU(2)^2U(1)} = \sum_{\text{doublets}} n_c Y = \tfrac{1}{2} - \tfrac{1}{2} = 0,$$

$$\mathcal{A}_{U(1)^3} = \sum n_c n_w Y^3 = \tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1 = 0,$$

$\mathcal{A}_{grav^2U(1)} = \sum n_c n_w Y = 1 - 2 + 1 - 1 + 1 = 0$ and $\mathcal{A}_{SU(3)^3} = 2 - 1 - 1 = 0$; three generations have twelve $SU(2)$ doublets (no Witten anomaly). Three lepton generations with a single quark generation would leave $\mathcal{A}_{SU(2)^2U(1)} = -1$ and $\mathcal{A}_{U(1)^3} = 3/2$.

The two halves of dirac16complex. B commutes with the four-dimensional Lorentz generators and with C, so $\Psi = \Psi_A + \Psi_B$ ($B = -1$, $B = +1$) splits into Lorentz-invariant halves and $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$; the identity $\Gamma_4 = -B\gamma^{x_0}$ (DERIVED, 16×16) gives $P_+ = (1 + \Gamma_4)/2 = (1 - B\gamma^{x_0})/2$, which equals $(1 + \gamma^{x_0})/2$ on Ψ_A and $(1 - \gamma^{x_0})/2$ on Ψ_B; the latter annihilates every orbital at the tip. The chiral charge densities $u^\dagger P_+ u$ at the tip (what a chiral tip current would couple to) are (DERIVED): Ψ_A fold band $a_0(-L)^2 = 0.163$ per unit length in each of its four blocks, Ψ_B fold band 0; Ψ_A odd (heavy) states 0.257, Ψ_B odd states 0; every transition current $\bar\Psi_A\gamma^\mu P_+\Psi_B$ vanishes at the tip.

The Z invisible width and the tip Z portal of version 1. Version 1 coupled the Ψ_A fold band (two Weyl species) to the Z through the chiral tip portal $S_{I9} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\,(c_Z + c'_Z\varsigma)(i\bar{\Psi}\gamma^\mu P_+\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu\Sigma\tau^3)$ ($\varsigma = \pm 1$ the side of the tip). With three neutrinos and a fold-band coupling r times a neutrino's,

$$N_\nu = \frac{\Gamma_{inv}}{\Gamma(Z\to\nu\bar{\nu})} = 3 + 2r^2;$$

the measured $N_\nu = 2.9963 \pm 0.0074$ (PDG 2025) is reproduced by $r = 0$ (0.5 standard deviations) and bounds $r^2 < 0.0056$ (two standard deviations). Even at that bound the neutral-current rate of the fold band, r^2 times the neutrino's ($\propto r^2 G_F^2 T^5$ against $H \propto T^2/M_{Pl}$), keeps it in equilibrium down to $T \approx 2.6\,\mathrm{MeV} \times r^{-2/3} = 14.7$ MeV, where $g_s = 10.75$, so it would carry the neutrino temperature and add

$$\Delta N_{eff} = n_W\Big(\frac{10.75}{g_s}\Big)^{4/3} F_h, \qquad F_h = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/(1+h_\nu)\Big]^4 = 1.0095$$

($n_W = 2$ Weyl species; F_h corrects for the non-instantaneous neutrino decoupling of step 5, $T_\nu/T_\gamma = 0.71664$ and the comoving heating $h_\nu = 0.00165$ with one common temperature: a species decoupled earlier is not heated by e^+e^- annihilation), $\Delta N_{eff} = 2.02$, far above the measured $N_{eff} = 2.99 \pm 0.17$ (Planck 2018) and 2.86 ± 0.13 (ACT DR6) (STANDARD). The oriented part c'_Z (odd under the fold reflection) would let the heavy odd Ψ_A states decay, $\Psi_A^{odd} \to \Psi_A^{fold} + Z$, and through the inverse decays and the exchange process $\Psi_A^{fold} Z \to \Psi_A^{fold} Z$ keep the fold band in equilibrium until the odd states disappear, $T \sim M_{DM}/30$ (above the electroweak scale, $g_s = 105.75$: the Standard Model without a Higgs particle), or, for c'_Z of the size of a neutrino's coupling, until the Z bosons annihilate near 15 GeV ($g_s = 86.25$: t, W, Z gone) (ESTIMATE): then $\Delta N_{eff} = 0.0958$ to 0.1257 and $N_{eff} = 3.139$ to 3.169, 2.1 to 2.4 standard deviations above ACT DR6 (the three-flavour value 3.0432: 1.4; disfavoured). UM-2 therefore drops the portal, $c_Z = c'_Z = 0$ (ASSUMED; the bounds DERIVED, `um2_electroweak.py`): $N_\nu = 3$, neither fold band is ever thermalised, and both halves keep an exact fold parity. (For reference, the plasma counts used are $g_s = 10.75$: photons, $e^\pm$, three neutrinos; 61.75 just above the QCD crossover: + gluons, u, d, s, μ; 86.25 below the top, W and Z masses; 105.75 above the electroweak scale. Right-handed neutrinos are never thermalised.)

The Z widths of three generations (tree level, massless fermions; $\Gamma(Z\to f\bar{f}) = N_c G_F m_Z^3(g_V^2 + g_A^2)/(6\sqrt{2}\pi)$, $g_V = T_3 - 2Q\sin^2\theta_W$ with the $\overline{MS}$ value $\sin^2\theta_W = 0.23122$, $g_A = T_3$; G_F, m_Z and the $\overline{MS}$ angle STANDARD; DERIVED):

channel	each ν	invisible (3ν)	each of e, μ, τ	each of u, c	each of d, s, b	hadronic	total
tree level (MeV)	165.9	497.7	83.4	285.4	367.9	1674.4	2422.3
with $1+\alpha_s/\pi$ on the quarks (MeV)						1737.3	2485.2
measured (PDG 2025, MeV)		499.1					2495.5

The top quark is too heavy for the Z to decay into it; $\alpha_s(m_Z) = 0.1180$ (STANDARD). The ratio $R_\ell = \Gamma_{had}/\Gamma_{\ell\ell} = 20.83$ compares with the measured 20.77 (STANDARD, PDG 2025). The remaining differences (0.4 % in the total) are the size of the electroweak loop corrections, which UM-2 does not compute; with the tree-level angle of the couplings of the mass fit, $\sin^2\theta_W = 0.22363$, the total changes by +0.82 % (a per-cent scheme ambiguity). The one-generation conflict of version 1 (a total of 1232 MeV) is gone.

8 Step 5: the flavour conversion of the three neutrinos

8.1 Masses, mixing and the equations

The neutrino masses are the x_0 eigenvalues of step 2 with the lightest mass chosen (ASSUMED) and the normal ordering (the NuFIT 6.0 best fit; the inverted ordering has $\Delta\chi^2 = 6.1$):

$$m_1 = 1.000 \text{ meV}, \qquad m_2 = \sqrt{m_1^2 + \Delta m_{21}^2} = 8.712 \text{ meV}, \qquad m_3 = \sqrt{m_1^2 + \Delta m_{31}^2} = 50.140 \text{ meV},$$

$\Delta m_{21}^2 = 7.49 \times 10^{-5}$ eV^2, $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ eV^2; $\sum m_\nu = 0.0599$ eV, below the DESI DR2 bound 0.064 eV (DERIVED, `um2_neutrino.py`, 16 checks). The PMNS matrix is the neutrino tip matrix of step 2. The flavour amplitudes ψ_α of a neutrino of energy E evolve with

$$i\frac{d\psi}{dx_4} = \mathcal{H}\psi, \qquad \mathcal{H} = \frac{1}{2E}\,U\,\mathrm{diag}(m_1^2, m_2^2, m_3^2)\,U^\dagger + \mathrm{diag}(V_{CC}, 0, 0), \qquad V_{CC} = \sqrt{2}\,G_F n_e$$

(the flavour-universal neutral-current potential drops out; antineutrinos: $U \to U^*$, $V_{CC} \to -V_{CC}$), solved exactly by $S = e^{-i\mathcal{H}L}$ for a constant density and by the ordered product of layer operators otherwise, with $P(\nu_\alpha \to \nu_\beta) = |S_{\beta\alpha}|^2$. In vacuum

$$P(\nu_\alpha \to \nu_\beta) = \Big|\sum_i U_{\beta i} U_{\alpha i}^* e^{-i m_i^2 L/(2E)}\Big|^2,$$

$$P(\nu_\mu \to \nu_e) - P(\nu_e \to \nu_\mu) = -16 J \sin\Delta_{21}\sin\Delta_{31}\sin\Delta_{32}, \qquad \Delta_{ij} = \frac{\Delta m_{ij}^2 L}{4E}.$$

8.2 Vacuum, matter and accelerators

Checks (DERIVED): the exact evolution reproduces the vacuum formula to 3.9×10^{-15}, unitarity to 1.8×10^{-15}, CPT ($P(\nu_\alpha \to \nu_\beta) = \bar{P}(\bar\nu_\beta \to \bar\nu_\alpha)$) to 2.8×10^{-16} and the T-odd difference to 5.9×10^{-14} at reactor, accelerator and atmospheric baselines. Reactor $\bar\nu_e$ of 4 MeV survive with $\bar{P}_{ee} = 0.9183$ at 1.65 km, 0.2263 at 52.5 km and 0.2674 at 180 km; the vacuum ν_μ survival has its first minimum, 0.0060, at $L/E = 1516$ km/GeV. The 1-3 MSW resonance, $2EV_{CC} = \Delta m_{ee}^2 \cos 2\theta_{13}$ with $\Delta m_{ee}^2 = c_{12}^2\Delta m_{31}^2 + s_{12}^2\Delta m_{32}^2$, lies at 11.135 GeV for $\rho = 2.8$ g/cm^3, $Y_e = 0.5$ ($n_e = \rho Y_e N_A$); the exact three-flavour Hamiltonian has $|U_{e3}^m|^2 = 1/2$ at 11.133 GeV (agreement 1.6×10^{-4}; with Δm_{31}^2 alone the formula gives 11.24 GeV). The 1-2 resonance at the centre of the Sun ($n_e = 100 N_A$ cm^{-3}), $2EV_{CC}c_{13}^2 = \Delta m_{21}^2 \cos 2\theta_{12}$, lies at 1.93 MeV. Long-baseline appearance in matter of constant density ($\delta = 212°$, normal ordering):

experiment	L (km)	E (GeV)	ρ (g/cm^3)	$P(\nu_\mu \to \nu_e)$	$\bar{P}(\bar\nu_\mu \to \bar\nu_e)$	vacuum $P(\nu_\mu \to \nu_e)$	$P(\nu_\mu \to \nu_\mu)$
T2K	295	0.6	2.6	0.0517	0.0313	0.0478	0.0085
NOvA	810	2.0	2.84	0.0481	0.0236	0.0404	0.1010
DUNE	1300	2.5	2.848	0.0701	0.0195	0.0491	0.0098

The matter effect raises the neutrino and lowers the antineutrino appearance for the normal ordering, and the phase $\delta = 212°$ adds to the asymmetry.

8.3 The Sun

With the electron density $n_e(r) = N_A \min(245\, e^{-10.54\, r/R_\odot}, 100)$ cm^{-3} (ASSUMED profile) and production at the radius r_0, the adiabatic survival probability is

$$P_{ee} = c_{13}^4\Big(\frac{1}{2} + \frac{1}{2}\cos 2\theta_{12}\cos 2\theta_{12}^m(r_0)\Big) + s_{13}^4, \qquad \cos 2\theta_{12}^m = \frac{\cos 2\theta_{12} - A}{\sqrt{(\cos 2\theta_{12} - A)^2 + \sin^2 2\theta_{12}}}, \qquad A = \frac{2EV_{CC}(r_0)c_{13}^2}{\Delta m_{21}^2},$$

and the numerical three-flavour evolution through 20000 and 40000 layers from r_0 to the surface, followed by the incoherent sum over the mass states, $P_{ee} = \sum_i |U_{ei}|^2 P(\nu_i)$, gives (DERIVED):

source	E (MeV)	$r_0/R_\odot$	P_{ee} adiabatic	P_{ee} numerical
pp	0.30	0.10	0.5409	0.5408
^{7}Be	0.862	0.06	0.5197	0.5196
pep	1.44	0.07	0.4978	0.4976
^{8}B	5.0	0.05	0.3771	0.3767
^{8}B	10.0	0.05	0.3194	0.3185
^{8}B	15.0	0.05	0.3055	0.3043

The two agree to 1.3×10^{-3} (the layer convergence is 2.2×10^{-6}): the conversion is adiabatic, vacuum dominated at low energy ($P_{ee}\to c_{13}^4(1-\frac{1}{2}\sin^2 2\theta_{12})+s_{13}^4$) and matter dominated for ^{8}B ($P_{ee}\to c_{13}^4\sin^2\theta_{12}+s_{13}^4$).

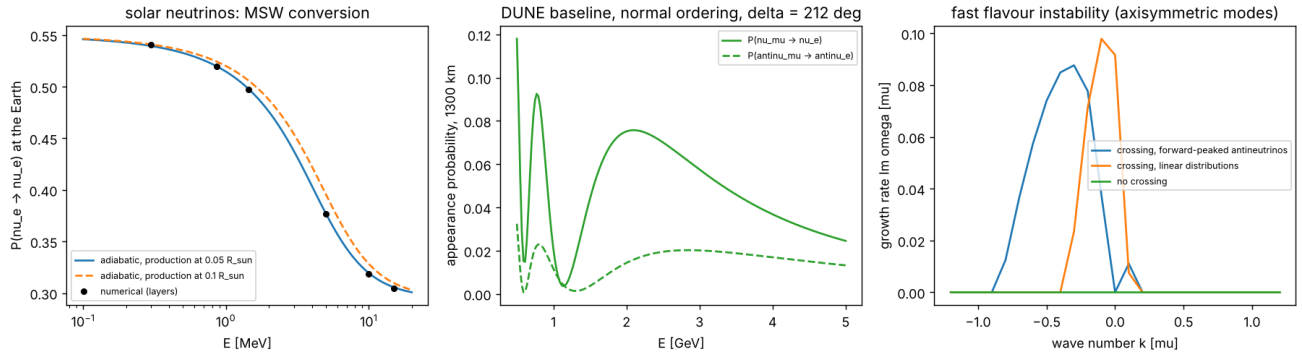


Figure 2: Flavour conversion in UM-2. Left: the solar ν_e survival probability (adiabatic formula for two production radii, numerical evolution as points). Middle: $\nu_\mu\to\nu_e$ and $\bar\nu_\mu\to\bar\nu_e$ appearance over 1300 km of rock (normal ordering, $\delta=212^\circ$). Right: the growth rate of the fast flavour instability against the wave number for the three angular distributions of the text.

8.4 The early Universe and N_{eff}

The thermal potential of the $e^\pm$ plasma,

$$V_e-V_\mu=-\frac{8\sqrt{2}\,G_F p}{3m_W^2}\,\frac{7\pi^2}{60}T^4,$$

falls below $\Delta m^2\cos2\theta/(2p)$ at the mean momentum $p=3.15T$ when $T=4.99$ MeV for the atmospheric and $T=2.39$ MeV for the solar splitting (DERIVED; m_W of the UM-2 fit): the flavours mix while the neutrinos decouple. The decoupling is computed from the temperature equations (Escudero's method; MeV units, the Hubble rate $H=\sqrt{8\pi\rho/3}/M_{Pl}$, $M_{Pl}=1.22091\times10^{19}$ GeV)

$$\frac{dT_\gamma}{dx_4}=-\frac{4H\rho_\gamma+3H(\rho_e+p_e)+3HT_\gamma\frac{dP_{int}}{dT_\gamma}+\frac{\delta\rho_{\nu_e}}{\delta x_4}+2\frac{\delta\rho_{\nu_\mu}}{\delta x_4}}{\frac{d\rho_\gamma}{dT_\gamma}+\frac{d\rho_e}{dT_\gamma}+T_\gamma\frac{d^2P_{int}}{dT_\gamma^2}},\qquad \frac{dT_{\nu_\alpha}}{dx_4}=-HT_{\nu_\alpha}+\frac{\delta\rho_{\nu_\alpha}/\delta x_4}{d\rho_{\nu_\alpha}/dT_{\nu_\alpha}},$$

$$\frac{\delta\rho_{\nu_e}}{\delta x_4}=\frac{G_F^2}{\pi^5}\Big[4(g_{eL}^2+g_{eR}^2)F(T_\gamma,T_{\nu_e})+2F(T_{\nu_\mu},T_{\nu_e})\Big],\qquad \frac{\delta\rho_{\nu_\mu}}{\delta x_4}=\frac{G_F^2}{\pi^5}\Big[4(g_{\mu L}^2+g_{\mu R}^2)F(T_\gamma,T_{\nu_\mu})-F(T_{\nu_\mu},T_{\nu_e})\Big],$$

$$F(T_1,T_2)=32f_a(T_1^9-T_2^9)+56f_sT_1^4T_2^4(T_1-T_2),\qquad f_a=0.884,\quad f_s=0.829,$$

with $\rho_\gamma=\pi^2T_\gamma^4/15$, $\rho_{\nu_\alpha}=7\pi^2T_{\nu_\alpha}^4/120$ per flavour (ν_τ as ν_μ), the $e^\pm$ density and pressure with the electron mass, the effective couplings $g_{eL}=0.727$, $g_{eR}=0.233$, $g_{\mu L}=-0.273$, $g_{\mu R}=0.233$ (or the UM-2 tree-level couplings $\frac{1}{2}+s^2$, s^2, $-\frac{1}{2}+s^2$, s^2, $s^2=0.22363$), and the QED interaction pressure $P_{int}=P_2+P_3$,

$$P_2=-\frac{e^2T^2}{12\pi^2}J_1-\frac{e^2}{8\pi^4}J_1^2+\frac{e^2m_e^2}{16\pi^4}J_2,\qquad P_3=\frac{e^3T}{12\pi^4}I^{3/2},\qquad \rho_{int}=-P_{int}+T\frac{dP_{int}}{dT},$$

$$J_1 = \int_0^\infty dp\,\frac{p^2}{E} n_D, \qquad J_2 = \int_0^\infty\!\int_0^\infty dp\,dq\,\frac{pq}{E_p E_q}\ln\Big|\frac{p+q}{p-q}\Big| n_D(p) n_D(q), \qquad I = \int_0^\infty dp\,\frac{p^2+E^2}{E} n_D,$$

$n_D = 2/(e^{E/T}+1)$, $E = \sqrt{p^2+m_e^2}$. The integrals are evaluated with Gauss-Legendre panels; the logarithmic singularity of J_2 at $q = p$ is removed by the substitution $q = p(1-s^2)$, and the massless limits $P_2 = -5e^2T^4/288$, $P_3 = e^3T^4/(36\sqrt{3}\pi)$, $\rho_e = 7\pi^2T^4/60$ are reproduced to 10^{-16}. The equations run from $T = 20$ MeV to 5 keV (LSODA, relative tolerance 10^{-11}), and

$$N_{eff} = \Big(\frac{11}{4}\Big)^{4/3}\Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4 + 2\Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4\Big];$$

for instantaneous decoupling at $T_d = 20$ MeV the plasma entropy $a^3(\rho+p)/T$ (with ρ_{int}, P_{int}) is conserved and $T_\nu \propto 1/a$. Results (DERIVED):

run	N_{eff}	published (Escudero 2020, same method, Table 1 there)
instantaneous decoupling, no QED	3.0002	3.000
instantaneous, $O(e^2)$ QED	3.0108	3.011
instantaneous, $O(e^2)+O(e^3)$	3.0098	3.010
Fermi-Dirac energy transfer, one neutrino temperature, no QED	3.0393	3.039
the same with $O(e^2)$	3.0495	
the same with $O(e^2)+O(e^3)$	3.0485	3.049
UM-2, separate ν_e and $\nu_{\mu,\tau}$ temperatures (no oscillations), $O(e^3)$	3.0476	3.048 (separate temperatures)
UM-2, separate temperatures, UM-2 tree-level couplings	3.0472	
UM-2, one common temperature (efficient oscillations)	3.0485	

With separate temperatures $T_{\nu_e}/T_\gamma = 0.71732$ and $T_{\nu_\mu}/T_\gamma = 0.71621$; e^+e^- annihilation raises the comoving neutrino temperatures by 0.265 % (ν_e, which also has charged-current interactions with the electrons) and 0.109 % ($\nu_{\mu,\tau}$). The plasma of UM-2 at MeV temperatures is that of the three-flavour Standard Model (the fold bands made by freeze-in, $\Delta N_{eff} = 0.001$ in version 3, step 9, are decoupled free-streaming radiation, which takes no part in the neutrino decoupling and is added to N_{eff} separately), so the precision Standard-Model value with oscillations and the electron mass in the rates is adopted:

$$N_{eff} = 3.0432 \pm 0.0002 \qquad \text{(Cielo, Escudero, Mangano and Pisanti 2023)},$$

from which the computed range 3.0476 to 3.0485 differs by 0.0044 to 0.0053 (the remaining difference is the electron mass in the rates and the momentum dependence, which the method of temperatures does not resolve).

8.5 The cosmic neutrino background

After decoupling the flavour coherence is lost: the CNB of UM-2 is an incoherent mixture of the three mass eigenstates with the Fermi-Dirac spectrum $f = 1/(e^{p/T_\nu}+1)$ ($T_{CMB} = 2.7255$ K, STANDARD),

$$T_\nu = \Big(\frac{4}{11}\Big)^{1/3} T_{CMB} = 1.9454\ \text{K} = 1.676\times10^{-4}\ \text{eV}, \qquad n_i = \frac{3\zeta(3)}{2\pi^2}T_\nu^3 = 112.0\ \text{cm}^{-3}\ (\nu_i+\bar\nu_i),$$

$$\rho_i = 2\int \frac{d^3p}{(2\pi)^3}\sqrt{p^2+m_i^2}\,f, \qquad \Omega_\nu h^2 = \frac{\sum_i \rho_i}{\rho_c/h^2} = 6.382\times10^{-4} \quad (\rho_c/h^2 = 1.05368\times10^4\ \text{eV/cm}^3),$$

compared with the non-relativistic value $\sum m_i/94.06$ eV $= 6.363\times10^{-4}$ at this temperature (the semi-relativistic ν_1, whose energy density exceeds $m_1 n_1$ by 16 %, adds 0.3 %), with the conventional $\sum m_i/93.14$ eV $= 6.426\times10^{-4}$

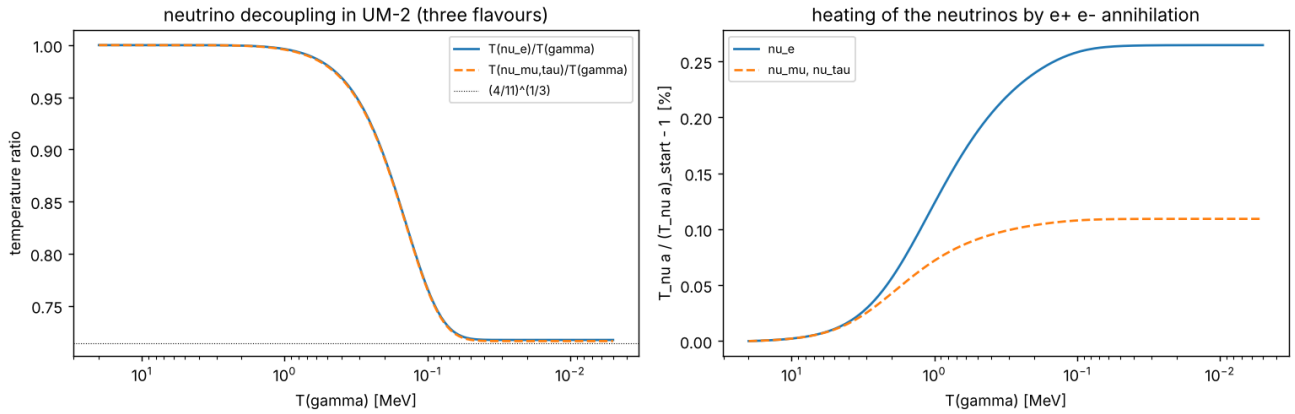


Figure 3: Neutrino decoupling in UM-2 (three flavours, separate ν_e and $\nu_{\mu,\tau}$ temperatures, QED to $O(e^3)$). Left: the neutrino temperatures relative to the photon temperature; right: the comoving heating of the neutrinos by e^+e^- annihilation.

(which includes the heating of the non-instantaneous decoupling) and with 6.451×10^{-4} for the heated spectrum of $N_{eff} = 3.0432$ (step 6: the degeneracy $N_{eff}/3$ of each state corresponds to a Fermi-Dirac spectrum of temperature $(N_{eff}/3)^{1/4} T_\nu$, so the non-relativistic density is multiplied by $(N_{eff}/3)^{3/4}$). The states (DERIVED; mean velocity $\langle v \rangle = \int p^2 (p/E) f \, dp / \int p^2 f \, dp$, $E = \sqrt{p^2 + m_i^2}$):

state	mass (meV)	n (cm^{-3})	$\langle v \rangle / c$ today	$\lvert U_{ei} \rvert^2$	$\lvert U_{\mu i} \rvert^2$	$\lvert U_{\tau i} \rvert^2$
ν_1	1.000	112.0	0.435	0.677	0.112	0.211
ν_2	8.712	112.0	0.0604	0.301	0.428	0.271
ν_3	50.140	112.0	0.0105	0.022	0.460	0.518

The fold bands of dirac16complex are not part of the CNB (they are never thermalised, step 4).

8.6 Supernovae

The MSW resonances of the envelope. With $\rho(r) = 10^4 \, (10^9 \, \mathrm{cm}/r)^3$ g/cm^3, $Y_e = 0.5$ (ASSUMED), a resonance lies at $n_e = \Delta m^2 \cos 2\theta / (2\sqrt{2} \, G_F E)$, i.e. at r with $d \ln n_e / dr = -3/r$, its adiabaticity is $\gamma = \Delta m^2 \sin^2 2\theta / (2E \cos 2\theta \, |d \ln n_e / dr|)$ and the jump probability $P_{LZ} = e^{-\pi\gamma/2}$ (DERIVED; two-flavour conditions: Δm^2_{31} and θ_{13} for the H resonance, Δm^2_{21} and θ_{12} for the L resonance, without the factor c^2_{13} of the solar condition, which would move the L densities by 2 %, and without Δm^2_{ee}, which would move the H densities by 1 %):

E (MeV)	H (1-3): ρ_{res} (g/cm^3)	r_{res} (km)	γ	L (1-2): ρ_{res} (g/cm^3)	r_{res} (km)	γ
10	3147	1.47×10^4	283	37.7	6.43×10^4	903
20	1573	1.85×10^4	178	18.8	8.10×10^4	569
40	787	2.33×10^4	112	9.42	1.02×10^5	358

Both resonances are adiabatic ($P_{LZ} \leq 2.6 \times 10^{-77}$). For the normal ordering a ν_e produced in the core leaves the star as ν_3, $P(\nu_e \to \nu_e) = |U_{e3}|^2 = 0.02215$, and an $\bar{\nu}_e$ as $\bar{\nu}_1$, $\bar{P}(\bar{\nu}_e \to \bar{\nu}_e) = |U_{e1}|^2 = 0.677$ (without collective effects and Earth matter). For representative accretion-phase neutrinospheres ($L = 6, 6, 4 \times 10^{52}$ erg/s and $\langle E \rangle = 11, 14, 15.5$ MeV for ν_e, $\bar{\nu}_e$, ν_x; ASSUMED), number fluxes $N \propto L / \langle E \rangle$ and the survival probabilities P, $\bar{P}$ above, the mean energies observed at the Earth,

$$\langle E \rangle_{\nu_e} = \frac{P N_e \langle E \rangle_e + (1-P) N_x \langle E \rangle_x}{P N_e + (1-P) N_x}, \qquad \langle E \rangle_{\bar{\nu}_e} = \frac{\bar{P} N_{\bar{e}} \langle E \rangle_{\bar{e}} + (1-\bar{P}) N_x \langle E \rangle_x}{\bar{P} N_{\bar{e}} + (1-\bar{P}) N_x},$$

become 15.29 MeV (ν_e) and 14.34 MeV ($\bar\nu_e$).

The fast flavour instability. With the ELN distribution $G(u) = \sqrt{2}G_F[n_{\nu_e}g_{\nu_e}(u) - n_{\bar\nu_e}g_{\bar\nu_e}(u)]$ in units of $\mu = \sqrt{2}G_F n_{\nu_e}$ ($G/\mu = g_{\nu_e} - \alpha g_{\bar\nu_e}$, $\alpha = n_{\bar\nu_e}/n_{\nu_e}$), the dispersion relation of the linearised equations for modes $e^{-i(\Omega x_4 - K x_3)}$ along the symmetry axis x_3, $\det\Pi = 0$ with $\Pi^{\mu\nu} = \eta^{\mu\nu} + \int du/2\, G\, v^\mu v^\nu/(\omega - ku)$, $v = (1, u)$, $\eta^{\mu\nu} = \mathrm{diag}(1, -1)$ on (x_4, x_3), has the axisymmetric branch $(1 + J_0)(-1 + J_2) - J_1^2 = 0$, $J_n = \int du/2\, G\, u^n/(\omega - ku)$, and the azimuthal branch $-1 + \int du/2\, G(1 - u^2)/(2(\omega - ku)) = 0$ ($\omega = \Omega - \Lambda^0$, $k = K - \Lambda^3$, $\Lambda^\mu = \int du/2\, G\, v^\mu$). The roots with $\mathrm{Im}\,\omega > 0$ were found for real k on a grid ($-1.2 \le k \le 1.2$, step 0.1) by vectorised Newton iterations with the analytic derivative from a grid of complex starting points (DERIVED):

angular distributions	ELN crossing	largest growth (branch, k)	homogeneous mode	mode $K = \Lambda^3$
$g_{\nu_e} \propto e^{-(1-u)^2/(2\cdot 0.6^2)}$, $g_{\bar\nu_e} \propto e^{-(1-u)^2/(2\cdot 0.4^2)}$, $\alpha = 0.8$	yes	$0.0877\,\mu$ (axisymmetric, $k = -0.30$)	$0.0185\,\mu$	stable
$g_{\nu_e} = 1 + 0.2u$, $g_{\bar\nu_e} = 1 + 0.6u$, $\alpha = 0.9$	yes	$0.0979\,\mu$ (axisymmetric, $k = -0.10$)	stable	$0.0917\,\mu$
as the first, $\alpha = 0.3$	no	none	stable	stable

(the largest azimuthal growth rates are $0.0200\,\mu$ and $0.0050\,\mu$ for the first two cases). Three independent evaluations agree to 1.2×10^{-14}: the analytic growth $\frac{1}{2}\sqrt{4I_1^2 - (I_0 + I_2)^2}$ ($I_n = \int du/2\, G\, u^n$) of the mode $K = \Lambda^3$ against the root of the dispersion relation at $k = 0$, and the eigenvalues of the discretised homogeneous equations $i\, dS_u/dx_4 = (\Lambda^0 - u\Lambda^3)S_u - \int du'/2\, G(u')(1 - uu')S_{u'}$ (300 nodes) against the root at $k = -\Lambda^3$. With an ELN crossing the gas is unstable, without one it is not, as Morinaga's theorem (2022) states; the second case shows that a stable homogeneous mode does not exclude growing inhomogeneous modes. Near the neutrinosphere, $n_{\nu_e} = 10^{32}$ to 10^{33} cm^{-3} ($\mu = 1.27 \times 10^{-5}$ to 1.27×10^{-4} eV), the first case grows by e in 5.9×10^{-10} to 5.9×10^{-11} s over 17.8 to 1.78 cm: a million times faster than the hydrodynamics, which is why explosion models treat the conversion as instantaneous.

Flavour conversion and the explosions. The charged-current heating behind the shock is, schematically, $q \propto N_e\langle E^2\rangle_e + N_{\bar e}\langle E^2\rangle_{\bar e}$ with $\langle E^2\rangle = \frac{5}{4}\langle E\rangle^2$ (spectra with the pinching parameter 3). Full three-flavour equipartition gives $q' = \frac{1}{3}(N_e\langle E^2\rangle_e + 2N_x\langle E^2\rangle_x) + \frac{1}{3}(N_{\bar e}\langle E^2\rangle_{\bar e} + 2N_x\langle E^2\rangle_x)$, and pair conversion $\nu_e\bar\nu_e \to \nu_x\bar\nu_x$ of $\Delta = \frac{2}{3}(N_{\bar e} - N_x)$ pairs, which conserves the ELN until $\bar\nu_e$ is equipartitioned, $q'' = (1 - \Delta/N_e)N_e\langle E^2\rangle_e + (1 - \Delta/N_{\bar e})N_{\bar e}\langle E^2\rangle_{\bar e}$; for the neutrinospheres above $q'/q = 0.884$ and $q''/q = 0.760$ (DERIVED, schematic): the heavy-lepton neutrinos are more energetic but fewer and do not heat through charged currents. Gogilashvili and Tamborra (2026, arXiv:2605.16504) simulated 195 progenitors of 9 to 120 solar masses (GR1D, M1 transport of ν_e, $\bar\nu_e$, ν_x in 18 energy bins, SFHo equation of state) with instantaneous flavour equipartition, conserving the electron lepton number and the number and momentum of the neutrinos in each energy bin, imposed at 0.02 s after bounce where the baryon density is below ρ_c (STANDARD, cited):

ρ_c (g/cm^3)	no conversion	10^9	10^{10}	10^{11}	10^{13}
failed explosions	25.6 %	50.8 %	77.4 %	93.3 %	96.4 %
weighted with the Salpeter mass function	27.0 %	47.6 %	61.9 %	73.9 %	88.3 %

The effect is strongest between 16 and 30 solar masses (easing the red-supergiant and supernova-rate problems), and the successful explosions leave lighter neutron stars (gravitational masses around 1.2 to 1.4 solar masses); the authors caution that the large fractions for deep conversion are in tension with observations and call for dynamical, multi-dimensional treatments.

8.7 Dirac or Majorana (version 3)

UM-2 gives the neutrinos Dirac masses (the lepton-doublet zero mode mixed at the tip with three right-handed singlets localised at $|c_R| = 31$ to 35, tip couplings of order one). The alternative without right-handed neutrinos is a Majorana mass from the gauge-invariant tip operator

$$S_M = -\frac{1}{2}\int_{y=-L} d^7x\sqrt{|g_{tip}|}\sum_{i,j} m_{ij}\,\overline{N_i^c}N_j + \text{h.c.}, \qquad N_i = (\Sigma^\dagger L_i)_1$$

(N^c the charge conjugate; the upper component of $\Sigma^\dagger L_i$ is neutral under $SU(2)_L \times U(1)_Y$ because $\Sigma \to U_L \Sigma U_Y^\dagger$), whose four-dimensional mass is, to first order in the tip value of the doublet zero mode,

$$m_\nu = \kappa_M \frac{\zeta(c_L)^2}{fL}, \qquad \zeta(c_L)^2 = \frac{c_L e^{-c_L}}{\sinh c_L} = 8.038\times 10^{-5}, \qquad \frac{1}{fL} = 22.1 \text{ TeV}$$

(DERIVED): the measured masses need $\kappa_M = 5.63\times10^{-13}$, 4.90×10^{-12}, 2.82×10^{-11}, a hierarchy that the x_0 mechanism does not explain, whereas the Dirac masses come from the localisation with couplings of order one: the mechanism favours Dirac neutrinos. Neutrinoless double-beta decay would have

$$m_{\beta\beta} = \Big|\sum_i U_{ei}^2\, m_i\, e^{i\alpha_i}\Big|, \qquad 0.84 \text{ meV} \le m_{\beta\beta} \le 4.41 \text{ meV},$$

for all Majorana phases α_i ($|U_{ei}|^2 m_i = 0.677$, 2.624, 1.111 meV; the extremes $\sum_i |U_{ei}|^2 m_i$ and $2\max_i - \sum_i$ agree with a scan of the phases), below the reach of present experiments; the non-relativistic CNB would be captured twice as often as for Dirac neutrinos (both helicities active); the flavour conversion is the same.

9 Step 6: the CMB, the CNB and the BAO

The four-dimensional effective theory is general relativity (step 1) with photons, baryons, the three neutrino mass eigenstates, the dark matter (the odd quanta of Ψ_A and Ψ_B), in version 3 the fold bands of dirac16complex made by freeze-in (massless free-streaming radiation with $\Delta N_{eff} = 0.001$, step 9, which obeys the massless limit $E_i = q$ of the neutrino hierarchy below) and Λ_4. Its linear perturbations obey the standard Einstein-Boltzmann equations (Newtonian gauge, conformal time, $ds^2 = a^2[-(1+2\psi)d\tau^2 + (1-2\phi)d\vec{x}^{\,2}]$):

$$k^2\phi + 3\mathcal{H}(\phi' + \mathcal{H}\psi) = -4\pi G_N a^2 \delta\rho, \qquad k^2(\phi' + \mathcal{H}\psi) = 4\pi G_N a^2(\bar\rho + \bar P)\theta,$$

$$\phi'' + \mathcal{H}(\psi' + 2\phi') + (2\mathcal{H}' + \mathcal{H}^2)\psi + \frac{k^2}{3}(\phi - \psi) = 4\pi G_N a^2 \delta P, \qquad k^2(\phi - \psi) = 12\pi G_N a^2(\bar\rho + \bar P)\sigma,$$

with the cold dark matter $\delta_c' = -\theta_c + 3\phi'$, $\theta_c' = -\mathcal{H}\theta_c + k^2\psi$; the baryons and photons coupled by Thomson scattering (n_e the free-electron density, σ_T the Thomson cross-section, $R = 4\bar\rho_\gamma/(3\bar\rho_b)$, c_s the baryon sound speed, $F_{\gamma l}$ and $G_{\gamma l}$ the temperature and polarisation multipoles, $F_{\gamma 2} = 2\sigma_\gamma$),

$$\delta_b' = -\theta_b + 3\phi', \qquad \theta_b' = -\mathcal{H}\theta_b + c_s^2 k^2\delta_b + R\,a n_e \sigma_T(\theta_\gamma - \theta_b) + k^2\psi,$$

$$\delta_\gamma' = -\tfrac{4}{3}\theta_\gamma + 4\phi', \qquad \theta_\gamma' = k^2\big(\tfrac{1}{4}\delta_\gamma - \sigma_\gamma\big) + k^2\psi + a n_e\sigma_T(\theta_b - \theta_\gamma),$$

$$F_{\gamma l}' = \frac{k}{2l+1}\big[lF_{\gamma(l-1)} - (l+1)F_{\gamma(l+1)}\big] - a n_e\sigma_T\Big[F_{\gamma l} - \frac{\delta_{l2}}{10}\big(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}\big)\Big] \quad (l \ge 2),$$

$$G_{\gamma l}' = \frac{k}{2l+1}\big[lG_{\gamma(l-1)} - (l+1)G_{\gamma(l+1)}\big] + a n_e\sigma_T\Big[-G_{\gamma l} + \frac{1}{2}\big(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}\big)\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big];$$

and one massive hierarchy per neutrino mass state i (comoving momentum q, energy $E_i = \sqrt{q^2 + a^2 m_i^2}$, multipoles $\mathcal{N}_l^{(i)}$ of the perturbed Fermi-Dirac distribution $\bar f_\nu$): $\mathcal{N}_0^{(i)\prime} = -\frac{qk}{E_i}\mathcal{N}_1^{(i)} - \phi'\frac{d\ln\bar f_\nu}{d\ln q}$, $\mathcal{N}_1^{(i)\prime} = \frac{qk}{3E_i}(\mathcal{N}_0^{(i)} - 2\mathcal{N}_2^{(i)}) - \frac{E_i k}{3q}\psi\frac{d\ln\bar f_\nu}{d\ln q}$, $\mathcal{N}_l^{(i)\prime} = \frac{qk}{(2l+1)E_i}[l\mathcal{N}_{l-1}^{(i)} - (l+1)\mathcal{N}_{l+1}^{(i)}]$ ($l \ge 2$). They were solved with CAMB 2.0.5 at the Planck 2018 values (STANDARD) $\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\theta_{MC} = 1.04092$, $\tau_{reio} = 0.0544$ (the optical depth to reionisation), $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$ (θ held fixed, so H_0 is an output; H_0 in km/s/Mpc). In UM-2 the three mass states are evolved separately, each with the degeneracy $N_{eff}/3 = 1.0144$ (the heated spectrum);

CAMB gives each the density deg $\times \rho_{FD}(m';T_\nu)$ at $T_\nu = (4/11)^{1/3}T_{CMB}$, so its masses are $m'_i = m_i \deg^{-1/4}$, which reproduce the late-time density $m_i n_\nu \deg^{3/4}$ of the heated spectrum; CAMB finds m'_i by inverting the full (relativistic) density from ω_ν and the mass fractions, which were therefore iterated (13 iterations) until the masses agree to 7.8×10^{-12} (DERIVED). The models are ΛCDM with the Planck 2018 parameters (one neutrino of 0.06 eV, $N_{eff} = 3.044$: the CAMB defaults), UM-2 (version 3: the three mass states and the freeze-in fold bands, $N_{eff} = 3.0432 + 0.0010 = 3.0442$), UM-2 without the freeze-in fold bands (the neutrino content of version 2, $N_{eff} = 3.0432$), and two variants with a thermalised fold band of dirac16complex instead (two Weyl species, which UM-2 does not contain; steps 4 and 9), decoupled above the electroweak scale ($\Delta N_{eff} = 0.0958$) or just above the QCD crossover ($\Delta N_{eff} = 2(10.75/61.75)^{4/3} \times 1.0095 = 0.1963$) (DERIVED, `um2_cosmology.py`, 8 checks). The BAO distances $D_M(z) = \int_0^z c\,dz'/H(z')$ (flat), $D_H = c/H(z)$, $D_V = (zD_M^2 D_H)^{1/3}$, in units of the drag-epoch sound horizon r_d, are compared with the DESI DR2 measurements (STANDARD, arXiv:2503.14738)

tracer	z	D_V/r_d	D_M/r_d	D_H/r_d	r_{MH}
BGS	0.295	7.942 ± 0.075			
LRG1	0.510		13.588 ± 0.167	21.863 ± 0.425	-0.459
LRG2	0.706		17.351 ± 0.177	19.455 ± 0.330	-0.404
LRG3+ELG1	0.934		21.576 ± 0.152	17.641 ± 0.193	-0.416
ELG2	1.321		27.601 ± 0.318	14.176 ± 0.221	-0.434
QSO	1.484		30.512 ± 0.760	12.817 ± 0.516	-0.500
Lyα	2.330		38.988 ± 0.531	8.632 ± 0.101	-0.431

through $\chi^2 = \sum_i (d_i - d_i^{th})^T C_i^{-1} (d_i - d_i^{th})$, $C_i = \begin{pmatrix} \sigma_M^2 & r_{MH}\sigma_M\sigma_H \\ r_{MH}\sigma_M\sigma_H & \sigma_H^2 \end{pmatrix}$, for the pairs $d_i = (D_M/r_d, D_H/r_d)$ of the anisotropic tracers (d_i^{th} the computed pair) and $(d - d^{th})^2/\sigma_V^2$ for BGS:

model	H_0	Ω_m	N_{eff}	Y_{He}	age (Gyr)	z_*	r_{drag} (Mpc)	BAO χ^2 (13 points)
ΛCDM	67.34	0.3154	3.044	0.24566	13.799	1089.91	147.103	32.55
UM-2 (version 3)	67.38	0.3150	3.0442	0.24567	13.796	1089.91	147.104	31.49
UM-2 without the freeze-in fold bands (version 2)	67.37	0.3151	3.0432	0.24566	13.797	1089.91	147.109	31.72
UM-2 with a thermalised fold band (electroweak)	68.48	0.3050	3.1390	0.24693	13.698	1090.00	146.630	15.43

model	H_0	Ω_m	N_{eff}	Y_{He}	age (Gyr)	z_*	r_{drag} (Mpc)	BAO χ^2 (13 points)
UM-2 with a thermalised fold band (QCD)	69.65	0.2948	3.2395	0.24824	13.596	1090.10	146.133	10.65

$\Omega_\nu h^2 = 6.4511 \times 10^{-4}$ in CAMB, equal to 6.4510×10^{-4} from the CNB of step 5 times $(N_{eff}/3)^{3/4}$; the mass sum 0.0599 eV lies below the DESI DR2 bound 0.064 eV, which would allow $m_1 < 4.1$ meV. The UM-2 TT and EE spectra agree with Planck ΛCDM to 2.5×10^{-4} and 2.7×10^{-4} ($30 \le \ell \le 2500$; for TT at most 0.007 of the cosmic variance $\sqrt{2/(2\ell+1)}$ of a single multipole; the binned full-sky cosmic-variance $\chi^2 = \sum_b (\bar{\delta}_b/\sigma_b)^2$ over bins b of $\Delta\ell = 50$, with $\bar{\delta}_b$ the mean relative deviation and $\sigma_b = \sqrt{2/\sum_{\ell\in b}(2\ell+1)}$, is 0.020; without the freeze-in fold bands 2.8×10^{-4}, 2.8×10^{-4} and 0.023): UM-2 reproduces the CMB of ΛCDM. By themselves the three mass states instead of one state of 60 meV move TT and EE by 2.6×10^{-4} and 2.3×10^{-4}, and $N_{eff} = 3.0432$ instead of 3.044 by 1.3×10^{-4}. The fold-band variants, at fixed Planck ω_b, ω_c, θ_*, move TT by up to 1.5 % and 3.1 % (0.8 and 1.5 cosmic-variance units of a single multipole; binned $\chi^2 = 293$ and 1208): they are not fits to the CMB, and they lie 2.1 and 2.9 standard deviations above the ACT DR6 value $N_{eff} = 2.86 \pm 0.13$ (UM-2: 1.4). The CNB enters through $N_{eff} = 3.0432$ and the three neutrino masses.

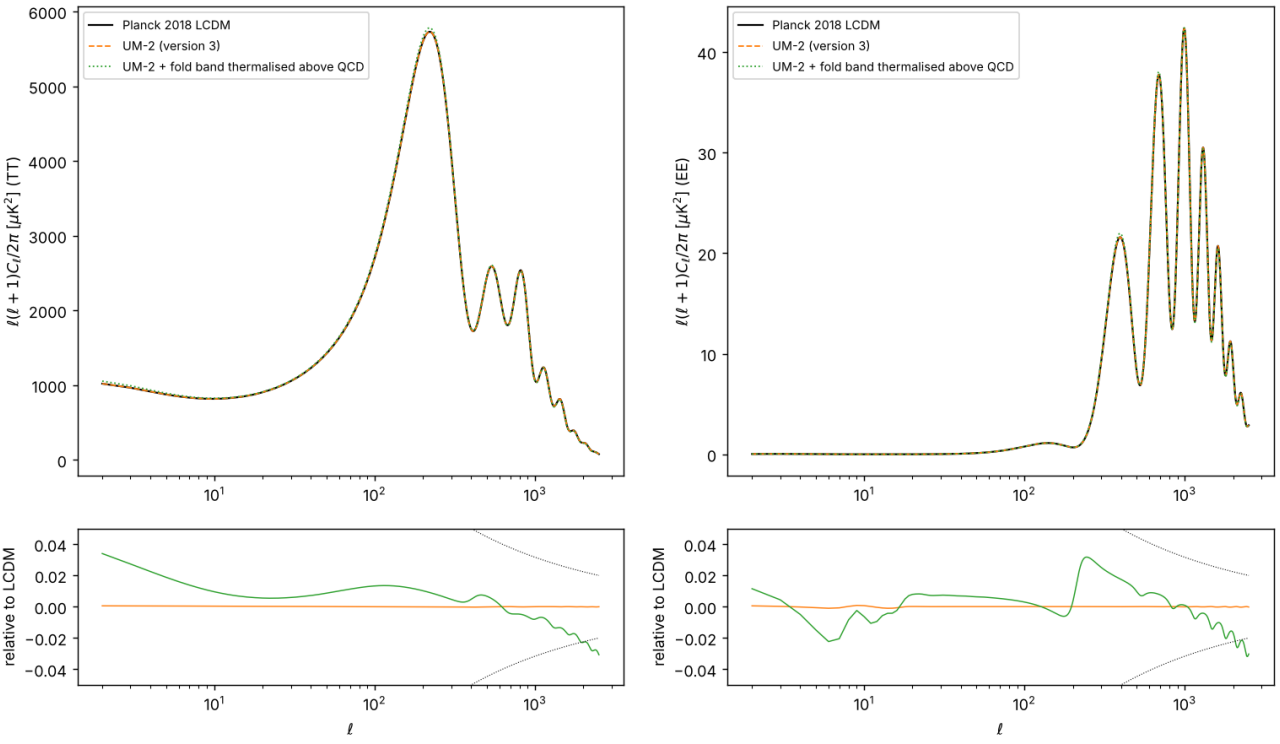


Figure 4: CMB temperature (left) and E-mode (right) spectra of UM-2 (three massive neutrinos), of Planck 2018 ΛCDM and of the variant with a fold band thermalised above the QCD crossover; lower panels: relative difference to ΛCDM with the cosmic variance of one multipole (dotted).

The matter power spectrum and version 3. A separate CAMB run of each model without the CMB gives

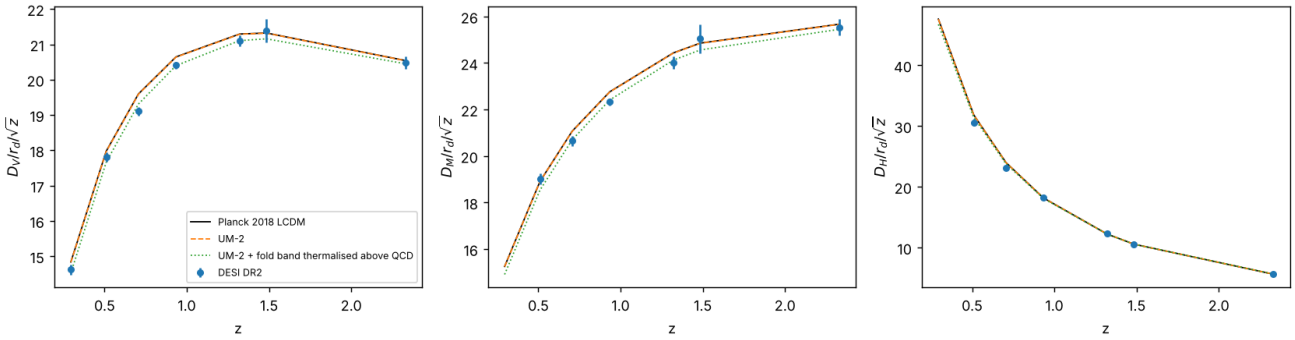


Figure 5: BAO distances divided by $\sqrt{z}$ against the DESI DR2 measurements; the χ^2 uses D_V for BGS and D_M, D_H with their correlation for the other tracers (the D_V points of the anisotropic tracers are shown for illustration).

the linear matter power spectrum $P_m(k)$ and

$$\sigma_8^2 = \int_0^\infty \frac{dk}{k}\,\frac{k^3 P_m(k)}{2\pi^2}\,W^2(kR), \qquad W(x) = \frac{3(\sin x - x\cos x)}{x^3}, \qquad R = 8\,h^{-1}\ \text{Mpc}:$$

$\sigma_8 = 0.8118$ (UM-2, version 3), 0.8119 (UM-2 without the freeze-in fold bands) and 0.8109 (ΛCDM; Planck 2018: 0.811, STANDARD); $f\sigma_8 = 0.4761$, 0.4762 and 0.4756 at $z = 0.5$ ($f = d\ln D/d\ln a$ the growth rate) (DERIVED). The fold bands produced by freeze-in through I11 (step 9) are massless free-streaming radiation with $\Delta N_{eff} = 0.001$: they move the TT spectrum by at most 1.6×10^{-4} against the model without them (DERIVED, 8 checks in all).

10 Step 7: the dark-matter relic and its detection

The lightest odd states of Ψ_A and of Ψ_B are two degenerate Dirac fermions of mass $M_{DM} = 3.102204\,m = 44.35$ TeV ($m = H$), cold and neutral; UM-2 has no tip operator, so neither couples to the Z, and both are stable by the fold parity (step 4). Through I12 they annihilate in the p-wave into gauge-boson pairs: for $\mathcal{L} = G\,\bar\chi\chi F_{\mu\nu}F^{\mu\nu}$ per abelian channel, $\sigma v = G^2 s(s-4M^2)/(4\pi) \to G^2M^4v^2/\pi$, in total $\sigma v = (M^4v^2/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$ (the polarisation sum $\sum|(k_1\cdot k_2)(e_1\cdot e_2) - (k_1\cdot e_2)(k_2\cdot e_1)|^2 = s^2/2$ and the spin trace $\mathrm{Tr}[(\not p_1 + M)(\not p_2 - M)] = 2(s-4M^2)$ checked numerically). Since $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$ does not mix the halves, each species freezes out separately and contributes half of $\Omega_c h^2 = 0.120$. Freeze-out (Dirac fermion, $g_* = 105.75$ at $T_f = M/x_f \approx 1.5$ TeV: the Standard Model of three generations without a Higgs particle; the right-handed neutrinos and the fold bands are never thermalised):

$$x_f = \ln\left[0.038\,c(c+2)\,g\,M_{Pl}M\langle\sigma v\rangle/\sqrt{g_* x_f}\right] = 30.38, \qquad \Omega h^2 = \frac{2\times1.07\times10^9\,\text{GeV}^{-1}\,x_f}{\sqrt{g_*}\,M_{Pl}\,3b/x_f} = 0.060 \text{ per species}$$

($\langle\sigma v\rangle = 6b/x$, $b = (M^4/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$, $c = 1/2$, $g = 2$, $M_{Pl} = G_N^{-1/2} = 1.22091\times10^{19}$ GeV, standard radiation-dominated freeze-out) requires $\langle\sigma v\rangle(x_f) = 2.01\times10^{-25}$ cm^3/s per species, i.e.

$$(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = (15.5\ \text{TeV})^{-3}, \qquad G_i = \mathcal{O}\,c_i g_i^2/\Lambda_\star^3, \quad \mathcal{O} = 0.7072$$

(a single species carrying all the dark matter would need $(17.6\ \text{TeV})^{-3}$ with the same g_*). These are the numbers of the freeze-out formula (version 2); the numerical Boltzmann solution of step 9 needs $(15.86\ \text{TeV})^{-3}$ at $x_f = 31.70$.

Unitarity (DERIVED, `um2_analysis.py`). The operator $\bar\chi\chi FF$ annihilates a fermion pair only from the $J = 0$, $P = +$ state ($^3P_0$); for unpolarised Dirac fermions the inelastic cross-section from it is bounded by $\sigma \le \pi/(4p^2)$, p the centre-of-mass momentum ($p = Mv/2$ for $v \ll 1$), i.e. $\sigma v \le \pi/(M^2 v)$, whose thermal average over the relative velocities, $f(v) \propto v^2 e^{-xv^2/4}$, is

$$\left\langle \frac{\pi}{M^2 v}\right\rangle = \frac{\sqrt{\pi x}}{M^2} = 5.79\times10^{-26}\ \text{cm}^3/\text{s} \quad \text{at } x_f = 30.38$$

(checked by numerical averaging). The freeze-out requirement exceeds it 3.47 times; the effective cross-section $G^2M^4v^2/\pi$ with $G = (8G_g^2 + 3G_W^2 + G_B^2)^{1/2}$ at the relic strength exceeds the bound for $v > v_c = (\pi/(GM^3))^{2/3} =$

0.263, i.e. for 79 % of the pairs at freeze-out (with the strength and x_f of the freeze-out formula; with the relic strength of step 9 the formula gives 0.274 and 75 % of the pairs, and the exact crossing lies at $v = 2\beta = 0.269$). (Version 1 compared the requirement with $(4\pi/M^2)\sqrt{x_f/\pi} = 2.32 \times 10^{-25}$ cm^3/s, the bound of a single helicity channel without the spin average and the restriction to $^3P_0$, a factor 4 too large.) The gauge-kinetic portal therefore cannot produce the relic abundance within perturbative unitarity. Step 9 recomputes this with the Boltzmann equation (a cross-section at the $^3P_0$ limit at every energy would leave $\Omega h^2 = 0.058$ per species: only a strongly coupled $^3P_0$ channel at the unitarity limit could give the relic) and resolves it in version 3 with the s-wave portal I13.

Direct detection (ESTIMATE). Even if the whole coupling were in gluons, the trace anomaly gives the nucleon coupling

$$f_N = G_g \frac{8\pi}{9\alpha_s} m_N f_{TG}, \qquad \langle N|G^\alpha G^\alpha|N\rangle = -\frac{8\pi}{9\alpha_s} m_N f_{TG}, \qquad \sigma_{SI} = \frac{\mu_N^2 f_N^2}{\pi}$$

($f_{TG} = 0.92$, $\alpha_s = 0.3$ at a low scale, $m_N = 0.939$ GeV, μ_N the dark-matter-nucleon reduced mass; FROM MEMORY), i.e. $\sigma_{SI} = 6.3 \times 10^{-53}$ cm^2 for each half (the two halves have the same cross-section and together the full local density, so the limit applies as for one species), eight orders below the LZ limit extrapolated linearly in the mass from 2.2×10^{-48} cm^2 at 40 GeV to 44 TeV (2.4×10^{-45} cm^2); the electroweak channels have no tree-level nucleon coupling. Any relic value would also depend on the radion oscillation left over from the deflation (which redshifts like dust): if it dominated the Universe until it decayed, it would dilute the dark matter after freeze-out (the moduli problem, below).

11 Step 8: the coupled solution in environments (version 3)

11.1 The portals on the zero modes

A four-dimensional particle of density n in an orbital or zero mode with the normalised profile $\varrho(y)$ ($\int_{-L}^{0} \varrho\, dy = 1$) has the proper eight-dimensional density $n\,\varrho(y)e^{-3Hy}/(2f^4 v_t)$ on each half of the period (v_t the coordinate volume of the extra times), so the portals I11 and I13 couple a state o of Ψ (orbital (a_o, b_o) normalised on one half, $\int_{-L}^{0}(a_o^2 + b_o^2)dy = 1$) to a chirality component f of a Standard-Model fermion (zero-mode profile $\varrho_c(y) = (2c/L)e^{2cy/L}/(1 - e^{-2c})$, localisation c) through

$$\mathcal{L}_4 = \big(G_V q_f \tilde{I}_{of} + G_A q_f^B \tilde{I}_{of}^{(b)}\big)(i\bar{\chi}_o\gamma^\mu\chi_o)(i\bar{f}\gamma_\mu f), \qquad \tilde{I}_{of} = \int_{-L}^{0}(a_o^2 + b_o^2)\varrho_f e^{-3Hy}dy, \qquad \tilde{I}_{of}^{(b)} = \int_{-L}^{0} b_o^2 \varrho_f e^{-3Hy}dy,$$

$G_V = c_V H/(2f^4 v_t \Lambda_\star^6)$, $G_A = c_A H/(2f^4 v_t \Lambda_\star^6)$, q_f the $B - L$ charge and $q_f^B = \frac{1}{3}$ for quarks, 0 for leptons; I13 acts on the lower component only, $h_j \to h_j + \frac{1}{2}(1 - \sigma_3)v_b(y)$, because $\gamma^{x_0} = \sigma_3$ in the block form. The overlaps (DERIVED; $H = 1$, 80 basis functions per sector):

zero mode f	Q	L	u_R	c_R	t_R	d_R	s_R	b_R	e_R	μ_R	τ_R	ν_{Ri}
$\tilde{I}$, dark matter	8.2722	1.2524	0.3854	0.6056	8.2722	0.3961	0.4620	0.7622	0.3972	0.5879	1.2524	0.302-0.308
$\tilde{I}^{(b)}$, dark matter	2.6135	0.9134	0.3795	0.5569	2.6135	0.3891	0.4459	0.6607	0.3900	0.5442	0.9134	0.301-0.308
$\tilde{I}$, fold band	6.685	1.659	0.792	1.060	6.685	0.807	0.893	1.221	0.808	1.041	1.659	0.668-0.678

zero mode f	Q	L	u_R	c_R	t_R	d_R	s_R	b_R	e_R	μ_R	τ_R	ν_{Ri}
$\tilde{I}$, first even bulk level	8.92	1.185	0.428	0.602	8.92	0.437	0.488	0.733	0.438	0.588	1.185	0.359-0.365

($\tilde{I}^{(b)} = 0$ for the fold band; ν_{Ri} the three right-handed neutrinos.) Checks (DERIVED): self-consistent Kohn-Sham states (eight fold zero modes and eight dark-matter quanta, $\lambda = 0$) with the mean field $e\,\varrho_f(y)e^{-3Hy}$ on both components (I11) or on the lower one (I13), at $e = \pm 10^{-4}$, $\pm 2\times 10^{-4}$ with Richardson extrapolation, reproduce $\tilde{I}$ for Q, e_R, L to 3.1×10^{-10} and $\tilde{I}^{(b)}$ for Q, L, u_R to 2.3×10^{-10}; with the benchmark self-coupling λ_B the response to the quark-doublet field is 8.2693 against 8.2722 (screening -3.5×10^{-4}); under the lepton-doublet I11 field the gap between the $k = 0$ fold zero mode and the first shell changes by -1.1×10^{-10} per unit amplitude (the second order, $-0.041e^2$, is a change of the group velocity: the ratio of the shell energies $|k| = 2\Delta k$ and Δk stays 2.000000000), and under I13 the fold levels move by less than 6.8×10^{-14}: the fold band stays massless and untouched by I13.

11.2 The mean fields of the Standard-Model fields

In an environment with the net densities n_p, n_n, n_e (unpolarised fermions, each chirality half; $n_u = 2n_p + n_n$, $n_d = n_p + 2n_n$) and the temperature T, the dark-matter level moves by

$$\delta\varepsilon_{I11} = G_V\Big[\tfrac{1}{3}n_u\tfrac{1}{2}(\tilde{I}_Q + \tilde{I}_{u_R}) + \tfrac{1}{3}n_d\tfrac{1}{2}(\tilde{I}_Q + \tilde{I}_{d_R}) - n_e\tfrac{1}{2}(\tilde{I}_L + \tilde{I}_{e_R})\Big],$$

$$\delta\varepsilon_{I13} = G_A\Big[\tfrac{1}{3}n_u\tfrac{1}{2}(\tilde{I}^{(b)}_Q + \tilde{I}^{(b)}_{u_R}) + \tfrac{1}{3}n_d\tfrac{1}{2}(\tilde{I}^{(b)}_Q + \tilde{I}^{(b)}_{d_R})\Big], \qquad \delta\varepsilon_{I12} = \mathcal{O}\,\frac{c}{\Lambda_\star^3}\Big[\langle g_s^2G^2\rangle + \langle g^2W^aW^a\rangle + \langle g'^2B^2\rangle\Big],$$

with the benchmark strengths $G_V = 4.38\times 10^{-17}$ GeV^{-2}, $G_A = 1.278\times 10^{-9}$ GeV^{-2}, $c/\Lambda_\star^3 = 1.106\times 10^{-14}$ GeV^{-3} (step 9) and $\mathcal{O} = 0.7072$. The condensates: from the trace anomaly

$$\Big\langle\frac{\alpha_s}{\pi}G^2\Big\rangle_T = \Big\langle\frac{\alpha_s}{\pi}G^2\Big\rangle_0 - \frac{8}{b}(\rho - 3p)_{QCD}, \qquad b = 11 - \frac{2}{3}n_f, \qquad \langle g_s^2G^2\rangle = 4\pi^2\Big\langle\frac{\alpha_s}{\pi}G^2\Big\rangle,$$

$\langle\frac{\alpha_s}{\pi}G^2\rangle_0 = 0.012$ GeV^4 (STANDARD), $(\rho - 3p)_{QCD}/T^4 = 4\exp[-\ln^2(T/0.2\text{ GeV})/0.5]$ below 1 GeV (lattice-like, ESTIMATE), $\frac{b}{3}(1 + \frac{5n_f}{12})\alpha_s^2$ above (α_s at the scale $2\pi T$, one loop; leading order: from $p = \frac{8\pi^2}{45}T^4[1 + \frac{21n_f}{32} - \frac{15}{4}(1 + \frac{5n_f}{12})\frac{\alpha_s}{\pi}]$ and $d\alpha_s/d\ln T = -\frac{b}{2\pi}\alpha_s^2$ through $\rho - 3p = T^5\,d(p/T^4)/dT$; the trace anomaly $\rho - 3p = -\frac{b\alpha_s}{8\pi}\langle G^2\rangle$ is written in the signature $(+,-,-,-)$, in which the trace of the energy-momentum tensor is $\rho - 3p$; in the signature $(+,+,+,-)$ of UM-2 both sides change sign and the relation above is the same). For the massive vector bosons the free part is $\langle F^2\rangle_T = -2(\rho - 3p)_V$ of their ideal gas (on shell the Proca Lagrangian, in the signature $(+,-,-,-)$ $-\frac{1}{4}F^2 + \frac{1}{2}m_V^2A_\mu A^\mu$, is a total derivative and $T^\mu{}_\mu = -m_V^2A_\mu A^\mu = \rho - 3p$; the result does not depend on the signature), and above m_W the order-α interaction pressure of the electroweak plasma dominates: by the Feynman-Hellmann theorem, for the Lagrangian $-F_{geo}^2/(4g^2)$ ($F_{geo} = gF$),

$$\langle F_{geo}^2\rangle = -4\,\frac{\partial p}{\partial(1/g^2)} = 4g^2p_2, \qquad p_2 = -\frac{g^2T^4}{144}\,d_A\Big[C_A + \frac{5}{2}T_FN_f\Big]$$

(STANDARD; $SU(2)$: $d_A = 3$, $C_A = 2$, $T_FN_f = 3$ for twelve Weyl doublets; $U(1)_Y$: $d_A = 1$, $C_A = 0$, $\frac{5}{2}T_FN_f \to \frac{5}{2}\sum_{Dirac}Y^2 = \frac{25}{2}$; no scalar term, and below m_W times the thermal suppression $\rho(m_W,T)/\rho(0,T)$ of a massive vector, ESTIMATE; for QCD, $d_A = 8$, $C_A = 3$, $T_F = \frac{1}{2}$, it reproduces the trace-anomaly condensate to 10^{-16}, DERIVED), with the couplings at $\mu = 2\pi T$ (one loop from the $\overline{MS}$ values at m_Z without a Higgs particle: $1/\alpha_i(\mu) = 1/\alpha_i(m_Z) + \frac{b_i}{2\pi}\ln(\mu/m_Z)$, $b_2 = \frac{10}{3}$, $b_Y = -\frac{20}{3}$, $1/\alpha_2(m_Z) = \hat{s}_Z^2/\alpha$, $1/\alpha_Y(m_Z) = (1 - \hat{s}_Z^2)/\alpha$, $\hat{s}_Z^2 = 0.23122$, $\alpha = 1/127.95$), so that

$$\langle g^2W^aW^a\rangle = g^2\big[2(-2I_W) + \cos^2\theta_W(-2I_Z)\big] + 4g^2p_2^{SU(2)},$$

$$\langle g'^2 B^2\rangle = g'^2 \sin^2\theta_W(-2I_Z) + 4g'^2 p_2^{U(1)}, \qquad I_V = (\rho - 3p)_V$$

(θ_W the angle of the couplings). The cold environments: the cosmic mean today (the Planck baryon density with the helium fraction $Y_p = 0.2457$, $\Omega_c h^2 = 0.120$), the Galaxy at the Sun ($n_p = n_e = 1$ cm^{-3}), the Earth (mean density 5.51 g/cm^3, 48 % protons), the centre of the Sun (150 g/cm^3, 66 % protons, 32 % neutrons), a white dwarf (10^6 g/cm^3 of carbon and oxygen, equal numbers of protons and neutrons) and a neutron-star core (0.3 fm^{-3}, 10 % protons, $n_e = n_p$), with the local dark-matter density 0.4 GeV/cm^3 in the bound objects. In the early-Universe environments the net baryon density is $\eta_b n_\gamma$ with $\eta_b = 6.1\times10^{-10}$, $n_\gamma = 2\zeta(3)T^3/\pi^2$, shared as $n_n/n_p = e^{-Q/T}$ ($Q = 1.293$ MeV), $n_e = n_p$; the dark matter is in equilibrium for $M/T < 40$ ($n = 8(MT/2\pi)^{3/2}e^{-M/T}$ for both species) and has its relic density scaled with the entropy afterwards; the freeze-out environment is at $T = M/x_f$ with the $x_f = 31.23$ of step 9. Results (DERIVED):

environment	n_p (cm^{-3})	$\delta\varepsilon_{I11}$ (GeV)	$\delta\varepsilon_{I13}$ (GeV)	$\delta\varepsilon_{I12}$ (GeV)	$\delta\varepsilon/M_{DM}$	$\delta\ln f$
cosmic mean today	2.2×10^{-7}	3×10^{-64}	4×10^{-57}	3.7×10^{-15}	8.4×10^{-20}	7×10^{-94}
the Galaxy at the Sun	1	1×10^{-57}	1.5×10^{-50}	3.7×10^{-15}	8.4×10^{-20}	3×10^{-87}
the Earth (5.51 g/cm^3)	1.6×10^{24}	4.4×10^{-33}	4.9×10^{-26}	3.7×10^{-15}	8.4×10^{-20}	1×10^{-62}
centre of the Sun (150 g/cm^3)	6.0×10^{25}	1.1×10^{-31}	1.3×10^{-24}	3.7×10^{-15}	8.4×10^{-20}	3×10^{-61}
white dwarf (10^6 g/cm^3)	3.0×10^{29}	7.9×10^{-28}	8.9×10^{-21}	3.7×10^{-15}	8.4×10^{-20}	2×10^{-57}
neutron-star core (0.3 fm^{-3})	3.0×10^{37}	4.3×10^{-19}	4.4×10^{-12}	3.7×10^{-15}	1.0×10^{-16}	9×10^{-49}
nucleosynthesis ($T = 1$ MeV)	1.5×10^{22}	2.4×10^{-35}	2.8×10^{-28}	3.7×10^{-15}	8.4×10^{-20}	7×10^{-60}
QCD crossover ($T = 200$ MeV)	7.8×10^{28}	2.0×10^{-28}	2.3×10^{-21}	1.9×10^{-15}	4.4×10^{-20}	3×10^{-48}
electroweak era ($T = 100$ GeV)	9.7×10^{36}	2.6×10^{-20}	2.8×10^{-13}	-2.4×10^{-6}	-5.5×10^{-11}	7×10^{-39}
dark-matter freeze-out ($T = 1.42$ TeV)	2.8×10^{40}	7.3×10^{-17}	8.2×10^{-10}	-6.8×10^{-2}	-1.5×10^{-6}	1.1×10^{-34}

The QCD vacuum condensate dominates at low temperature ($\langle g_s^2 G^2\rangle = 0.474$ GeV^4, half of it left at 200 MeV); the thermal condensates dominate in the electroweak era and at the freeze-out, where the shift reaches 1.5×10^{-6} of the mass ($\langle g_s^2 G^2\rangle = -8.1\times10^{12}$, $\langle g^2 W^a W^a\rangle = -5.0\times10^{11}$, almost all from the order-α interaction, $\langle g'^2 B^2\rangle = -2.6\times10^{10}$ GeV^4).

11.3 dirac16complex acting on the Standard Model, and gravity

Through I12 the scalar density of the dark matter changes the gauge couplings, $\delta\alpha_i/\alpha_i = -4G_i n_{DM}$: -1.7×10^{-16} for α_s at the freeze-out, -1.1×10^{-20} in the electroweak era, at most -1.1×10^{-36} from nucleosynthesis on, -1.7×10^{-60} in the Galaxy today. In the x_0 problem of the W with the kinetic function $K = 1 + e\,s_{DM}(y)$ ($s_{DM} = 2ab$ of the dark-matter orbital),

$$-\frac{1}{Ke^{3\eta\xi}}\big(Ke^{3\eta\xi}u'\big)' = \Lambda u, \qquad u'(0) = 0, \qquad K(-1)\,u'(-1) = \beta_W u(-1),$$

quadratic finite elements (300 elements, Richardson in e) give $d\ln\Lambda_W/de = -0.1126$, while the photon and gluon zero mode keeps $\Lambda = 0$ (-2.5×10^{-11} at $e = 0.01$, the discretisation level); the physical amplitude, $e \sim 4c\,g^2 n_{DM}/\Lambda_\star^3 = |\delta\alpha/\alpha|$, is 1.7×10^{-16} at the freeze-out, 1.1×10^{-20} in the electroweak era, at most 1.1×10^{-36} from nucleosynthesis on and 5×10^{-66} for the cosmic mean today.

Gravity: the block responds to the block-averaged sources of every species with the adiabatic response of step 1,

$$\delta k = \big(0.035852\,\kappa\delta\rho - 0.107556\,\kappa\delta p_3 - 0.140028\,\kappa\delta\bar{p}\big)/H^2, \qquad \delta\ln f = -\frac{\delta k}{2k_*}, \qquad \delta\ln m_s = \frac{d\ln m_s}{d\ln f}\,\delta\ln f,$$

in which radiation drops out, with $\kappa\delta\rho = 8\pi G_N G(k_*)\rho_4$ for a four-dimensional density ρ_4 (likewise p_3 and $\bar{p}$). The hidden pressure is the response of the pressure to the block scale at fixed temperature, $\bar{p} = \frac{1}{4}\,\partial p/\partial\ln f$:

$$\bar{p} = -\frac{1}{4}\sum_s(\rho_s - 3p_s)\frac{d\ln m_s}{d\ln f} - \frac{1}{4}(\rho-3p)_{QCD}\frac{d\ln\Lambda^{(n_f)}}{d\ln f} + \frac{1}{4}\sum_{i=2,Y} p_2^{(i)}\,\frac{d\ln\alpha_i}{d\ln f},$$

the species of mass $m_s(f)$ as ideal gases with their x_0 masses (for an ideal gas $m\,\partial p/\partial m = -(\rho-3p)$; non-relativistic matter gives $-\frac{1}{4}\rho\,d\ln m/d\ln f$), the net nucleons and electrons as degenerate Fermi gases in the cold environments, the QCD plasma through $\partial p/\partial\ln\Lambda = -(\rho-3p)$ (for $p = T^4\varphi(T/\Lambda)$) and the electroweak plasma through its order-α pressure with $d\ln\alpha_i(\mu)/d\ln f = -\alpha_i(\mu)[4/\alpha_i(m_{KK}) + b_i/(2\pi)]$ (the bulk couplings scale as $\alpha_i(m_{KK}) \propto f^{-4}$ and $m_{KK}\propto 1/f$); in the trace the QCD and electroweak plasmas add $(\rho-3p)_{QCD}$ and $-\frac{b_i\alpha_i}{2\pi}p_2^{(i)}$. The sensitivities: $d\ln m_e/d\ln f = -20.5$, $d\ln M_{DM}/d\ln f = -0.772$, for a fermion of mass $m = e/(fL)$ with the first-order root $e \simeq \kappa_f\zeta(c_L)\zeta(-c_R)$ of step 2 and localisations $c = f\mu L \propto f$

$$\frac{d\ln m}{d\ln f} = -1 + \frac{1}{2}\Big[\frac{d\ln\zeta(c_L)^2}{d\ln c_L} + \frac{d\ln\zeta(-c_R)^2}{d\ln(-c_R)}\Big] = -18.82,\ -18.01,\ -14.90,\ -12.15,\ -10.89,\ -6.77$$

for u, d, s, c, b, t, and for the nucleon, through the QCD scale ($\alpha_s(m_{KK}) = 0.0606$ from $1/\alpha_s(m_{KK}) = 1/\alpha_s(m_Z) + \frac{7}{2\pi}\ln(m_{KK}/m_Z)$, $\alpha_s\propto f^{-4}$, $m_{KK}\propto 1/f$, $\Lambda^{(6)} = m_{KK}e^{-2\pi/(7\alpha_s(m_{KK}))}$, $\Lambda^{(3)} = \Lambda^{(6)\,21/27}(m_c m_b m_t)^{2/27}$, $\sigma_{\pi N} = 45$ MeV, $\sigma_s = 40$ MeV)

$$\frac{d\ln\Lambda^{(6)}}{d\ln f} = -1 - \frac{8\pi}{7\alpha_s(m_{KK})} = -60.2, \qquad \frac{d\ln\Lambda^{(3)}}{d\ln f} = \frac{21}{27}\frac{d\ln\Lambda^{(6)}}{d\ln f} + \frac{2}{27}\sum_{q=c,b,t}\frac{d\ln m_q}{d\ln f} = -49.0,$$

$$\frac{d\ln\Lambda^{(5)}}{d\ln f} = \frac{21}{23}\frac{d\ln\Lambda^{(6)}}{d\ln f} + \frac{2}{23}\frac{d\ln m_t}{d\ln f} = -55.6, \qquad \frac{d\ln\Lambda^{(4)}}{d\ln f} = \frac{21}{25}\frac{d\ln\Lambda^{(6)}}{d\ln f} + \frac{2}{25}\Big(\frac{d\ln m_t}{d\ln f} + \frac{d\ln m_b}{d\ln f}\Big) = -52.0,$$

$$\frac{d\ln m_p}{d\ln f} = \Big(1 - \frac{\sigma_{\pi N}+\sigma_s}{m_p}\Big)\frac{d\ln\Lambda^{(3)}}{d\ln f} + \frac{\sigma_{\pi N}}{2m_p}\Big(\frac{d\ln m_u}{d\ln f} + \frac{d\ln m_d}{d\ln f}\Big) + \frac{\sigma_s}{m_p}\frac{d\ln m_s}{d\ln f} = -46.1$$

(ESTIMATE, one loop). The block moves by at most $\delta\ln f = 1.1\times10^{-34}$, at the freeze-out, where the hot plasma has $\bar{p} = 3.1\times10^{12}$ GeV^4 (mostly the QCD interaction measure with $d\ln\Lambda/d\ln f = -60$) and $\rho - 3p_3 = 2.2\times10^{11}$ GeV^4 (the nucleon mass moves by 5.3×10^{-33}); 7×10^{-39} in the electroweak era; at most 9×10^{-49} from nucleosynthesis to today. The gain of the loop mass → source → block → mass is at most 1.1×10^{-33}, so the iteration converges after one step in every environment. (Version 3 as first written kept only the hidden pressure of non-relativistic matter and only the $\delta\bar{p}$ term of the response and found 5.6×10^{-45}; the conclusion is unchanged.) **The coupled solution is therefore the stabilised background with the x_0 spectra and the Kohn-Sham states of steps 1 to 3, corrected by the mean fields of this step; the corrections are computed and below 1.5×10^{-6}.** The I12 benchmark changes the strong coupling by 10 % through the thermal dark-matter density only at $T = 27.4$ TeV ($M/T = 1.62$), so the coupled effective description holds through the freeze-out and everything later (DERIVED, 16 checks with step 9).

12 Step 9: the thermal history and the dark-matter relic (version 3)

12.1 The thermal history

The species in equilibrium (photon, gluons, W^+, W^-, Z, six quarks, three charged leptons, three neutrinos; no Higgs particle; not the right-handed neutrinos, the fold bands, the radion or the Kaluza-Klein modes; the dark matter is not counted, since it would add less than 0.1 at 5 TeV and nothing at its freeze-out) with their x_0 masses give

$$\rho_i = \frac{g_i}{2\pi^2}\int_0^\infty \frac{p^2E\,dp}{e^{E/T}\pm1}, \qquad p_i = \frac{g_i}{6\pi^2}\int_0^\infty \frac{p^4\,dp}{E(e^{E/T}\pm1)}, \qquad s_i = \frac{\rho_i + p_i}{T}, \qquad g_\rho = \frac{30\sum_i\rho_i}{\pi^2T^4}, \quad g_s = \frac{45\sum_i s_i}{2\pi^2T^3},$$

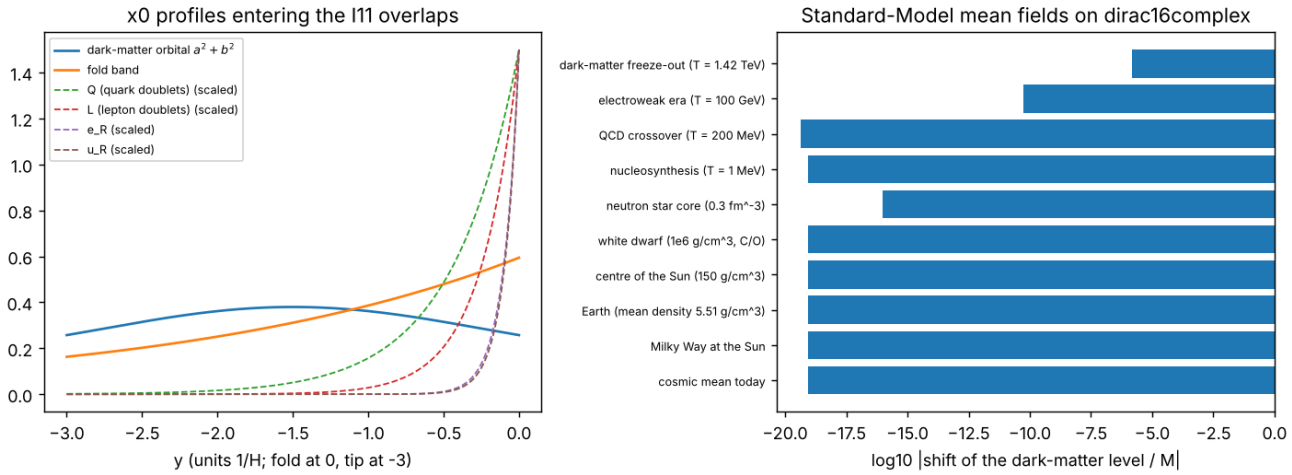


Figure 6: The coupled solution: the x_0 profiles that enter the overlaps (left) and the shift of the dark-matter level by the Standard-Model mean fields in the ten environments (right).

with the QCD crossover modelled by the weight $w = \frac{1}{2}[1 + \tanh((T - 155\ \text{MeV})/15\ \text{MeV})]$ of the quark-gluon plasma against a gas of the twelve lightest hadron multiplets (ASSUMED; masses in GeV and internal states: π 0.1380, 3; K 0.4956, 4; η 0.5479, 1; ρ 0.7753, 9; ω 0.7827, 3; K^* 0.8917, 12; N 0.9389, 8; η' 0.9578, 1; ϕ 1.0195, 3; Λ 1.1157, 4; Σ 1.1932, 12; Δ 1.232, 32; from memory) and neutrinos decoupled at 2 MeV ($T_\nu/T = (g_{s,e\gamma}/(11/2))^{1/3}$ afterwards) (DERIVED):

T	5 TeV	1.42 TeV	200 GeV	10 GeV	1 GeV	300 MeV	155 MeV	100 MeV	3 MeV	1 MeV	0.1 MeV	today
g_ρ	105.75	105.73	104.95	86.22	76.34	64.57	42.35	17.54	10.74	10.56	4.30	3.363
g_s	105.75	105.73	104.68	86.13	75.50	64.14	41.76	17.16	10.74	10.56	4.78	3.909

i.e. 105.75 above the top threshold (three generations without a Higgs particle), 86.25 without t, W, Z, 10.75 before $e^\pm$ annihilation, and today $g_s = 43/11 = 3.9091$, $g_\rho = 3.3626$, $s_0 = 2891.3\ \text{cm}^{-3}$ ($T_0 = 2.7255$ K).

12.2 The Boltzmann equation

For one Dirac species ($Y = (n_\chi + n_{\bar\chi})/s$, $x = M/T$, $s = \frac{2\pi^2}{45} g_s T^3$)

$$\frac{dY}{dx} = -\frac{(1+\delta)\, s\, \langle\sigma v\rangle}{2xH}(Y^2 - Y_{eq}^2), \qquad \delta = \frac{1}{3}\frac{d\ln g_s}{d\ln T}, \qquad H = \left(\frac{8\pi^3 g_\rho}{90}\right)^{1/2}\frac{T^2}{M_{Pl}}, \qquad Y_{eq} = \frac{4M^2 T K_2(x)}{2\pi^2 s},$$

$$\langle\sigma v\rangle = \frac{1}{8M^4 T K_2(x)^2}\int_{4M^2}^{\infty} \sigma(s)\,(s-4M^2)\sqrt{s}\,K_1(\sqrt{s}/T)\,ds, \qquad \Omega h^2 = \frac{M Y_\infty s_0}{\rho_c/h^2}$$

(K_1, K_2 the modified Bessel functions of the second kind; $\rho_c/h^2 = 1.05368\times10^{-5}$ GeV/cm^3; implicit Radau integration of $\ln Y$ from $x = 5$ to 2000, relative tolerance 10^{-10}; the thermal average by adaptive quadrature in the kinetic energy of the pair in units of its rest energy, $s/(4M^2)-1$, with exponentially scaled Bessel functions, tabulated on 260 points in x). Validation (DERIVED): an s-wave cross-section of 4.4×10^{-26} cm^3/s gives $\Omega h^2 = 0.1275$ (the freeze-out formula 0.1306), and the late-time integral $1/Y_\infty - 1/Y(x_*) = \int_{x_*}(1+\delta)s\langle\sigma v\rangle/(2xH)\,dx$ holds to 1.3×10^{-8}; for I12, $\sigma = G^2 s^{3/2}(s-4M^2)^{1/2}/(8\pi)$, the exact average exceeds the non-relativistic $6b/x$ ($b = G^2M^4/\pi$) by 0.408, 0.111, 0.031, 0.0030 at $x = 10$, 30, 100, 1000, i.e. by c_1/x with $c_1 \approx 3$.

12.3 I12 alone, and partial-wave unitarity

The effective operator would need $G = (8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = 2.509\times10^{-13}\ \text{GeV}^{-3} = (15.86\ \text{TeV})^{-3}$ for $\Omega h^2 = 0.060$ per species ($x_f = 31.70$, $\langle\sigma v\rangle(x_f) = 1.89\times10^{-25}$ cm^3/s; step 7: $(15.5\ \text{TeV})^{-3}$ from the freeze-out

formula). Its cross-section exceeds the ${}^3P_0$ bound $\sigma \le \pi/(4p^2)$ above $\sqrt{s} = 89510$ GeV (815 GeV above threshold), i.e. above the relative velocity $v = 2\beta = 0.269$ (0.274 from the non-relativistic formula $v_c = (\pi/(GM^3))^{2/3}$); at freeze-out 96 % of the annihilation rate comes from collisions above the crossing (the fraction of the Gondolo-Gelmini integral above it), and 75 % of the pairs of the non-relativistic Maxwell distribution are faster than 0.274. With $\sigma = \pi/(4p^2)$ at every energy, the largest annihilation from ${}^3P_0$, each species leaves $\Omega h^2 = 0.0582$ (at its freeze-out, $x_f = 30.77$, $\langle\sigma v\rangle = 5.33 \times 10^{-26}$ cm^3/s against the non-relativistic $\sqrt{\pi x}/M^2 = 5.84 \times 10^{-26}$ cm^3/s at the same x); the effective operator capped at the bound leaves 0.1544, 0.1081, 0.0836, 0.0649, 0.0585 for $G^2/G^2_{relic} = 1$, 3, 10, 100, 10^4. Within perturbative couplings I12 therefore overproduces the dark matter; only a strongly coupled ${}^3P_0$ channel at the unitarity limit could give the relic. Version 3 sets I12 to the unitarity-safe benchmark

$$G = \frac{\pi}{2M^3} = 1.801 \times 10^{-14}\ \text{GeV}^{-3} = (38.2\ \text{TeV})^{-3},$$

below the bound up to $\sqrt{s} = 106363$ GeV $= 2.40M$ (a fraction 5.2×10^{-4} of the annihilation rate at $x = 30$ comes from above); with universal coefficients and the one-loop couplings at $2M = 88.7$ TeV ($1/\alpha_s = 1/\alpha_s(m_Z) + \frac{7}{2\pi}\ln(2M/m_Z)$, $1/\alpha_2 = \hat{s}^2_Z/\alpha + \frac{10}{3}\frac{1}{2\pi}\ln(2M/m_Z)$, $1/\alpha_1 = (1-\hat{s}^2_Z)/\alpha - \frac{20}{3}\frac{1}{2\pi}\ln(2M/m_Z)$ with the $\overline{MS}$ value $\hat{s}^2_Z = \sin^2\hat{\theta}_W(m_Z) = 0.23122$, $\alpha(m_Z) = 1/127.95$; $g_s^2 = 0.7786$, $g^2 = 0.3781$, $g'^2 = 0.1380$), $c/\Lambda^3_\star = G/(\mathcal{O}\sqrt{8g_s^4 + 3g^4 + g'^4}) = 1.106 \times 10^{-14}$ GeV^{-3} ($\Lambda_\star/c^{1/3} = 44.9$ TeV). Alone it would leave $\Omega h^2 = 8.0$ per species.

12.4 The relic with I13

With $G_f = G_A \frac{1}{3}\tilde{I}^{(b)}_f$ for each quark Weyl component ($\sum_f N_c(q_f\tilde{I}^{(b)}_f)^2 = 16.351$) the s-wave cross-section and its unitarity bound are

$$\sigma_{I13} = \sum_f \frac{N_c G_f^2\, s\,(1 + 2M^2/s)}{24\pi\beta}, \qquad \beta = \sqrt{1 - \frac{4M^2}{s}}, \qquad \sigma_{I13} \le \frac{3\pi}{4p^2} \quad (J = 1,\ {}^3S_1),$$

and the Boltzmann equation with $\sigma_{I12} + \sigma_{I13}$ gives each species $\Omega h^2 = 0.060$ for (DERIVED)

$$G_A = 1.278 \times 10^{-9}\ \text{GeV}^{-2} = 1.1 \times 10^{-4}\, G_F \qquad (x_f = 31.23),$$

with $\langle\sigma v\rangle(x_f) = 9.50 \times 10^{-26}$ cm^3/s from I13 and 9.9×10^{-28} cm^3/s from I12, the I13 cross-section capped at its bound, which it reaches only above $v = 0.51$ (22 % of the freeze-out rate then comes from collisions at the bound; without the cap $G_A = 1.269 \times 10^{-9}$ GeV^{-2}, and the uncapped cross-section exceeds the bound above $v = 0.52$, for 25 % of the rate). The relic of UM-2 version 3 is a thermal freeze-out with most annihilations (78 % of the rate) below the unitarity bound; the contact form of I13 is an effective description (for the quark doublet $G_f^{-1/2} = (G_A q_f \tilde{I}^{(b)}_Q)^{-1/2} = 30$ TeV, below the energies $\sqrt{s} \approx 89$ TeV of the annihilations), whose completion by a mediator is OPEN.

Why the baryon current. With the $B - L$ current instead ($\sum_f N_c(q_f\tilde{I}^{(b)}_f)^2 = 22.92$ over the quarks, the leptons and the ν_R, so the relic needs $G_A^{B-L} = G_A(16.351/22.92)^{1/2} = 1.072 \times 10^{-9}$ GeV^{-2} with the uncapped G_A; the three ν_R carry 1.2 % of the annihilation) the right-handed neutrinos would be thermalised: per flavour

$$\frac{dn_{\nu_R}}{dx_4} + 3Hn_{\nu_R} = \gamma_\nu\Big(1 - \frac{n^2_{\nu_R}}{n^2_{\nu_R,eq}}\Big), \qquad \gamma_\nu = \sum_{A,B} n^2_{\chi,eq}\frac{G_\nu^2 M^2}{2\pi}, \qquad n_{\nu_R,eq} = \frac{3}{4}\frac{\zeta(3)}{\pi^2}T^3, \qquad n_{\chi,eq} = 2\Big(\frac{MT}{2\pi}\Big)^{3/2}e^{-x},$$

$G_\nu = G_A^{B-L}\tilde{I}^{(b)}_{\nu_R}$ (one representative flavour, the rms of the three overlaps, which differ by 2 %; $G_\nu^2M^2/(2\pi)$ the s-wave σv into one Weyl ν_R pair); from zero abundance at $M/T = 2$ they reach 0.9997 of the equilibrium number by $M/T = 40$; the number-changing rate falls below the Hubble rate at $M/T = 16.1$, and elastic scattering on the dark matter, $\Gamma_{el} = \sum_{A,B} 2n_{\chi,eq}\,\sigma_{el}$ with $\sigma_{el} = G_\nu^2E^2/(2\pi)$ for a Weyl ν_R of energy $E \simeq 3T$ on a heavy Dirac fermion (ESTIMATE), at $M/T = 24.4$ ($T = 1816$ GeV, $g_s = 105.7$); the three ν_R would add $\Delta N_{eff} = 3(10.75/g_s)^{4/3}F_h = 0.144$ ($F_h = 1.0095$), $N_{eff} = 3.187$, 2.5 standard deviations above ACT DR6 (2.86 ± 0.13). With J^B the ν_R keep their tip coupling and the I11 coupling of charge -1, through which the dark matter annihilates into ν_R pairs at only 2.0×10^{-17} of the I13 rate into quarks (the ratio $G_V^2\sum_i \tilde{I}^2_{\nu_{Ri}}/(G_A^2\sum_f N_c(q_f\tilde{I}^{(b)}_f)^2)$, DERIVED): they are never thermalised.

12.5 The fold bands: freeze-in through I11

With $G_f = G_V q_f \tilde{I}_f^{fold}$ the pair production $f\bar{f} \to \Psi_{fold}\bar{\Psi}_{fold}$ ($\sigma = G_f^2 s/(12\pi)$ per Dirac f, spin averaged; half for one chirality) has the rate density and energy injection (Maxwell-Boltzmann; $\int_0^\infty x^6 K_1 = 384$, $\int_0^\infty x^7 K_2 = 3072$, both checked)

$$\gamma = \frac{4}{\pi^5} N_\chi G_V^2 T^8 \sum_f N_c (q_f \tilde{I}_f)^2, \qquad \frac{d(\rho_{fold} a^4)}{dx_4} = 8T\gamma\, a^4, \qquad a \propto \frac{g_s^{-1/3}}{T}, \qquad \Delta N_{eff} = \frac{\rho_{fold}}{\frac{7}{8}\frac{\pi^2}{15} T_\nu^4},$$

$N_\chi = 2$, $\sum_f N_c (q_f \tilde{I}_f)^2 = 126.85$ over the Weyl components of three generations (no ν_R in the plasma). Integrated from T_{max} (DERIVED): $\Delta N_{eff} = 5.21 \times 10^{29}\, G_V^2$ GeV^4 for $T_{max} = 10$ TeV, so the benchmark $\Delta N_{eff} = 0.001$ means $G_V = 4.38 \times 10^{-17}$ GeV^{-2}; for that G_V, $\Delta N_{eff} = 8.0 \times 10^{-6}$, 1.25×10^{-4}, 1.0×10^{-3}, 8.0×10^{-3} at $T_{max} = 2$, 5, 10, 20 TeV, and at T_{max} the production rate per fold quantum is 0.012 of the Hubble rate: never in equilibrium.

12.6 The self-interaction channel (version 3, after the review)

The contact self-interaction $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ contains $-\lambda(\bar{\chi}_D\chi_F)(\bar{\chi}_F\chi_D)\, s_{DF}(y)^2$, with the transition scalar density $s_{DF} = a_D b_F + b_D a_F$ of the dark-matter orbital (a_D, b_D) and the fold zero mode $(a_F, 0)$; s_{DF} is odd under the fold reflection and its square even, so the dark matter can annihilate into the fold bands, with the four-dimensional strength (normalised as I11)

$$G_\lambda = \frac{\lambda H \tilde{I}_\lambda}{2 f^4 v_t}, \qquad \tilde{I}_\lambda = \int_{-L}^{0} s_{DF}^2\, e^{-3Hy}\, dy = 14.38 \text{ (units of } H),$$

and the s-wave cross-section estimated as $\sigma v = G_\lambda^2 M^2/(2\pi)$ per final fold species, $N_F = 2$ (the value of one vector channel; the Fierz coefficients are not computed: ESTIMATE). While the dark matter is in equilibrium ($T_{max} = 10$ TeV to $x_f = 31.23$) each annihilation puts $2M$ into the fold bands,

$$\frac{d(\rho_{fold} a^4)}{dx_4} = 2M\, \gamma_\lambda\, a^4, \qquad \gamma_\lambda = \sum_{A,B} n_{\chi,eq} n_{\bar{\chi},eq}\, \sigma v\, N_F, \qquad n_{\chi,eq} = 2\Big(\frac{MT}{2\pi}\Big)^{3/2} e^{-M/T},$$

which gives (DERIVED) $\Delta N_{eff} = 1.83 \times 10^{26}\, G_\lambda^2$ GeV^4 as long as the fold bands stay out of equilibrium. $\Delta N_{eff} \le 0.001$ therefore requires $G_\lambda \le 2.3 \times 10^{-15}$ GeV^{-2} ($1.8 \times 10^{-6}\, G_A$; the production rate per fold quantum, $2\gamma_\lambda / n_{fold,eq}$ with $n_{fold,eq} = N_F \cdot 4 \cdot \frac{3}{4}\frac{\zeta(3)}{\pi^2} T^3$, is then at most 0.027 of the Hubble rate). The self-coupling $\lambda_B = 7 \times 10^{-5} H^{-6}$ of the Kohn-Sham tests of step 3 would, with $v_t = H^{-3}$, give $G_\lambda = 1.14 \times 10^{-9}$ GeV^{-2}, keep the fold bands in equilibrium down to $T = 2.6$ TeV (production rate up to 6.3×10^9 times the Hubble rate) and add $\Delta N_{eff} = 4(10.75/g_s)^{4/3} F_h = 0.19$ (four Weyl species, $N_{eff} = 3.23$). UM-2 therefore ASSUMES $G_\lambda \le 2.3 \times 10^{-15}$ GeV^{-2}, i.e. $\lambda \le 1.4 \times 10^{-10} H^{-6}$ for $v_t = H^{-3}$, or $v_t \ge 4.9 \times 10^5 H^{-3}$ for $\lambda = \lambda_B$; with such a λ the self-interaction mean fields of every Kohn-Sham state of the cosmology are negligible, and λ_B is only the value with which the Kohn-Sham machinery was tested.

12.7 Detection

Per nucleon, with $G_{u,d} = G_A \frac{1}{3}\frac{1}{2}(\tilde{I}_Q^{(b)} + \tilde{I}_{u_R,d_R}^{(b)}) + G_V \frac{1}{3}\frac{1}{2}(\tilde{I}_Q + \tilde{I}_{u_R,d_R})$,

$$f_p = 2G_u + G_d, \qquad f_n = G_u + 2G_d, \qquad f_N = \frac{Z f_p + (A-Z) f_n}{A}, \qquad \sigma_n = \frac{\mu_N^2 f_N^2}{\pi}$$

(μ_N the dark-matter-nucleon reduced mass; xenon $Z = 54$, $A = 131$): $f_p = 1.915 \times 10^{-9}$, $f_n = 1.917 \times 10^{-9}$ GeV^{-2}, $\sigma_n = 4.01 \times 10^{-46}$ cm^2, 0.16 of the LZ limit linearly extrapolated to 44.35 TeV (2.44×10^{-45} cm^2; ESTIMATE): allowed and within reach of the next generation of xenon detectors (I11 alone: 3.9×10^{-60} cm^2). The s-wave annihilation into quark pairs continues today with $\sigma v = \sum_f N_c G_f^2 M^2/(2\pi) = 9.76 \times 10^{-26}$ cm^3/s (DERIVED, 7 + 16 checks).

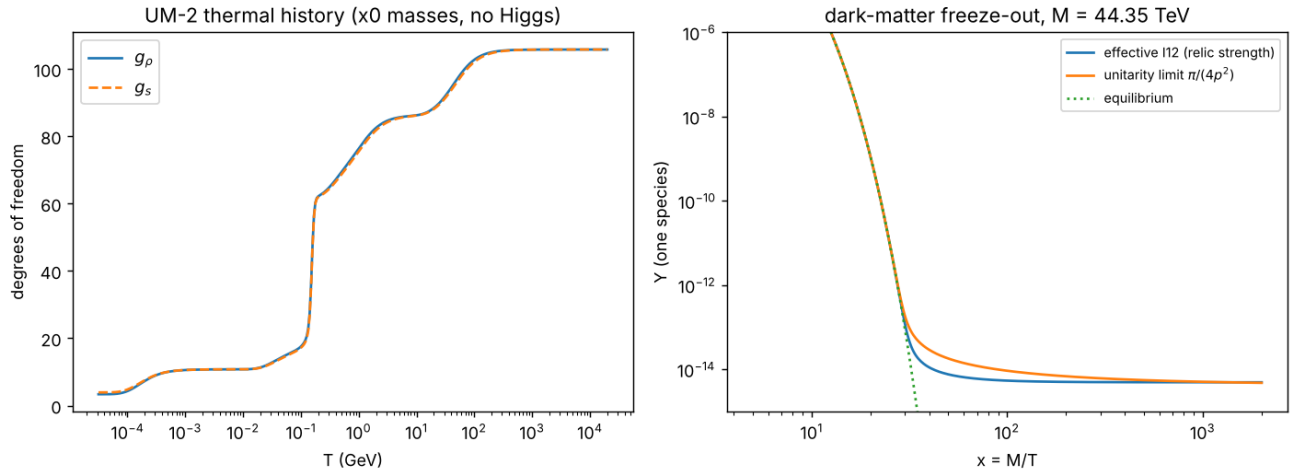


Figure 7: The thermal history of UM-2 (degrees of freedom with the x_0 masses, left) and the freeze-out of one dark-matter species through the effective I12 at the relic strength and at the ${}^3P_0$ unitarity limit (right).

13 Step 10: the terms of the metric power series (version 3)

The solution fixes the magnitude of every term of $g_{MN} = \bar{g}_{MN} + \epsilon h^{(1)}_{MN} + \epsilon^2 h^{(2)}_{MN} + \ldots$, $h^{(1)}_{MN} = \sum_n h^{(1,n)}_{MN}(x^\mu) u_n(x_0)$ (DERIVED, 4 checks):

- order 0, the background $\bar{g}$: 1;
- order 1, $n = 0$: the rms Newtonian potential per logarithmic wavenumber today, $\Delta_\Phi = [k^3 P_\Phi/(2\pi^2)]^{1/2}$ from the Weyl potential $\frac{1}{2}(\phi + \psi)$ of step 6 (UM-2, version 3): 2.42×10^{-5}, 2.28×10^{-5}, 1.51×10^{-5}, 4.12×10^{-6}, 1.74×10^{-6}, 3.55×10^{-7}, 5.03×10^{-8} at $k = 10^{-4}$, 10^{-3}, 10^{-2}, 0.05, 0.1, 0.3, 1 Mpc^{-1}; the Poisson equation $\Phi = \frac{3}{2}\Omega_m H_0^2 \delta_m/(ak^2)$ applied to the matter density agrees to 1.5×10^{-6} for $k \geq 0.01$ Mpc^{-1}, and on the largest scales $\frac{3}{5} A_s^{1/2} (k/k_0)^{(n_s-1)/2} g(0)$ ($k_0 = 0.05$ Mpc^{-1}, $g(0) = D(1) = 0.7878$ with $D(a) = \frac{5}{2}\Omega_m E(a) \int_0^a db/[bE(b)]^3$, $E(a) = (\Omega_m a^{-3} + 1 - \Omega_m)^{1/2}$) to 4.4×10^{-5}; bound objects $\Phi = G_N M/(Rc^2)$: Earth 7.0×10^{-10}, Sun 2.1×10^{-6}, white dwarf 1.0×10^{-4}, neutron star 0.17, galaxies $(v/c)^2 = 5.4 \times 10^{-7}$, clusters 1.1×10^{-5};
- order 1, $n \geq 1$: the graviton tower ($\beta = 0$) $u_n = e^{-a\xi}(\cos q_n \xi + \frac{a}{q_n} \sin q_n \xi)$, $q_n = n\pi$, $a = 3\eta/2$, $\Lambda_n = 9\eta^2/4 + n^2\pi^2 = 30.1196$, 59.7284, 109.0764 (finite elements agree to 3.9×10^{-9}), $N_n = \int_{-1}^0 e^{3\eta\xi} u_n^2 d\xi = \frac{1}{2}(1 + a^2/q_n^2)$ (to 1.8×10^{-10}), $N_0 = (1 - e^{-3\eta})/(3\eta)$; the potential of a static source $V = -\frac{G_N M m}{r}[1 + \sum_n \alpha_n e^{-m_n r}]$ with $\alpha_n = \frac{N_0}{N_n}(\int \varrho_M u_n \, d\xi)(\int \varrho_m u_n \, d\xi)$ for the profiles ϱ_M, ϱ_m of the two masses, i.e. for two masses with the same profile $\frac{N_0}{N_n}(\int \varrho u_n \, d\xi)^2$: 0.0728, 0.1469, 0.1809 at the fold, 0.0904, 0.0136, 0.0057 for the quark-doublet profile, 0.0208, 0.0197, 0.0105 for the lepton doublet, 0 for the flat gauge profile (to 1.7×10^{-10}); $m_1 = m_{KK} = 121.32$ TeV, range 1.63×10^{-21} m, $e^{-m_1 r} = e^{-3.1 \times 10^{16}}$ at 50 micrometres;
- the radion: $\delta \ln f \leq 1.1 \times 10^{-34}$ (step 8), range 4.08×10^{-21} m, the primordial oscillation decayed (lifetime $0.43\,\mathrm{s}/C_r$);
- order 2 (ESTIMATE, the square of order 1): $\Delta_\Phi^2 = (2.42 \times 10^{-5})^2 = 5.8 \times 10^{-10}$ on cosmological scales ($k = 10^{-4}$ Mpc^{-1}), 4.5×10^{-12} at the Sun, 0.03 at a neutron star;
- tensor modes: not predicted; $r < 0.036$ (from memory) bounds $\Delta_h < 8.7 \times 10^{-6}$.

On cosmological scales the terms decrease as 1, 2.4×10^{-5}, 5.8×10^{-10}, 10^{-34} (radion), 0 (Kaluza-Klein): the metric is a power series of terms of decreasing magnitude whose $n = 0$ first-order term carries the CMB, the CNB and the BAO.

14 Accuracy of the solution

- Exact (symbolic): the background equations, the static block, $G(k_*)$, the radion frequency, the first-order block response, the dimensional reduction, the field reductions and the zero modes.
- Numerical, with measured errors: boson levels (mpmath, 30 digits; finite elements $\leq 5 \times 10^{-10}$); fermion masses (exact roots; the first-order formula within 1.1×10^{-3} for the top quark and 3×10^{-4} for every other fermion; shooting $\leq 6.1 \times 10^{-10}$); the three-generation tip determinant (mpmath, 50 digits; agreement with the one-generation roots 3.0×10^{-42}); Kohn-Sham energies (Richardson errors below 2.2×10^{-9} relative; conservation 9×10^{-8}); primordial integrations (constraint below 6×10^{-10}); flavour evolution (vacuum formula 3.9×10^{-15}; solar layers converged to 2.2×10^{-6}; the 1-3 resonance formula 1.6×10^{-4}); neutrino decoupling (LSODA, relative tolerance 10^{-11}; the method reproduces the published values to 10^{-3}); fast flavour growth rates (1.2×10^{-14} between three evaluations); CMB (CAMB default accuracy, lensing included; the three neutrino masses to 7.8×10^{-12}). Version 3: the portal overlaps against self-consistent Kohn-Sham derivatives (3×10^{-10}); the Boltzmann solver against its late-time integral (1.3×10^{-8}) and the freeze-out formula (2.4 %); the relic strengths to 10^{-6} in $\ln G$ (root finding); the graviton tower against finite elements (4×10^{-9}); the potentials from the Weyl variable against the Poisson equation (1.5×10^{-6}); σ_8 at the CAMB default accuracy; after the review, the algebra of the I13 bilinear (exact), the Feynman-Hellmann pressure against the trace anomaly (10^{-16}) and the rate fractions above the unitarity crossings by adaptive quadrature with the breakpoints of the thermal weight.
- Approximations: the static block with an assumed stabiliser; the good sector (no extra-time momenta); the Kohn-Sham mean-field treatment (local exchange of the contact interaction); portal couplings as external mean fields; tree level for the electroweak widths; the coherent (collisionless) flavour evolution in the Sun, the Earth and the supernova envelope, with schematic density profiles; neutrino decoupling with one temperature per flavour (momentum averaged); the linear analysis of the fast flavour instability and a schematic estimate of its effect on the heating; the freeze-out formula (step 7; replaced by the Boltzmann equation in step 9); order ϵ for the cosmology. Version 3: ideal gases with a modelled QCD crossover and instantaneous neutrino decoupling for the degrees of freedom; Maxwell-Boltzmann statistics for the freeze-in; the effective cross-sections capped at the partial-wave bounds (a completion that saturates them); the QCD interaction measure below 1 GeV from a lattice-like shape and leading-order perturbation theory above; the order-α electroweak pressure without scalars; one-loop running for the couplings and the proton sensitivity; the elastic ν_R rate and the self-interaction cross-section as estimates.

15 What the solution does not reproduce, and what is open

- **Conflicts with data, inherent in the order**: no Higgs boson (one is observed at 125 GeV), and without it the longitudinal W^+W^- scattering violates perturbative unitarity near $\sqrt{8\pi}\, v \approx 1.2$ TeV, far below the Kaluza-Klein modes. (The one-generation conflict of version 1, a total Z width of 1.23 GeV, is removed: three generations give 2.485 GeV against the measured 2.4955 GeV.)
- **Open**: the origin of the stabiliser; the extra-time modes outside the good sector (tachyonic for momentum $q > fM$); the closed time-like curves implied by compact extra times; the radion left over from the deflation (ESTIMATE: a Planck-suppressed decay $\Gamma = C_r\, m_r^3/(4\pi \bar{M}_{Pl}^2)$, with the reduced Planck mass $\bar{M}_{Pl} = 2.435 \times 10^{18}$ GeV and an order-one coupling factor C_r, gives the lifetime $0.43\,\mathrm{s}/C_r$ and the reheating temperature $T_{RH} = (90/(\pi^2 g_*))^{1/4} \sqrt{\Gamma \bar{M}_{Pl}} = 1.85\sqrt{C_r}$ MeV, $g_* = 10.75$, below the 4.7 MeV that nucleosynthesis needs unless $C_r > 6.5$: a moduli problem); the block-shape mode driven by $p_0 - p_t$; the tuning of Λ_4 ($10^{-92} H^2$); the Kaluza-Klein graviton spectrum with the Lovelock corrections; the electroweak S parameter of the Higgsless tip and the electroweak loop corrections; anomalies of the eight-dimensional bulk theory; the absolute neutrino mass scale and the mass ordering (bounded and preferred by data, chosen here); the collective flavour evolution of supernova neutrinos beyond the linear analysis; the joint CMB and BAO fit of the fold-band variants; the microscopic origin of the portals (I12 alone could give the relic only as a strongly coupled ${}^3P_0$ channel; version 3 obtains it through I13, whose contact form needs a mediator above about 30 TeV); the self-coupling λ (bounded by ΔN_{eff}, ASSUMED); the highest temperature of the hot Universe ($T_{max} = 10$ TeV assumed for the freeze-in); the order-two terms of the metric in detail; the primordial spectrum, an input.